Generalized Linear Models

MTH303: Linear Statistical Models

Thanks to Dr. Mu He for teaching me GLM. I find GLM a remarkably charismatic theory: one idea, a linear predictor joined to an exponential family through a link, builds the foundation of modern statistics and of computer-age statistical inference, and the same ideas carry over into neural networks and AI.

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Part I. Preliminaries

1 Linear algebra

Where this is going

Teacher's road map on the first page: statistical learning in R/Python; computer-age statistical inference (Efron); stochastic processes (Hao Zhang); nonlinear statistical theory in signal processing.

The course covers: a review of preliminary mathematics (linear algebra, calculus, probability, statistics); an introduction to linear statistical models; linear regression models (estimation, inference, diagnostics, R code); maximum likelihood methods; generalized linear models (estimation, inference, diagnostics, R code); Bayesian statistics and its estimator.

1.1 Vectors and matrices

An n-dimensional column vector and an n-by-m matrix are

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \ (n \times 1), \qquad \boldsymbol{A} = \boldsymbol{A}_{n,m} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{pmatrix} \ (n \times m).$$

A matrix is a collection of vectors of the same dimension; a column vector is an n-by-1 matrix; a matrix is *square* if the numbers of rows and columns are equal.

1.2 Matrix operations

(a) **Transpose:** interchange rows and columns, $\boldsymbol{A}^\top = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{n1} \\ a_{12} & a_{22} & \cdots & a_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1m} & a_{2m} & \cdots & a_{nm} \end{pmatrix}$, $\boldsymbol{x}^\top = (x_1, \ldots, x_n)$. The transpose of an n-by-m matrix is m-by-n. $\boldsymbol{A}$ is *symmetric* if $\boldsymbol{A} = \boldsymbol{A}^\top$ (only possible for a square matrix).

(b) **Addition/subtraction:** for the same shape, element-wise, $(\boldsymbol{A} \pm \boldsymbol{B})_{ij} = a_{ij} \pm b_{ij}$.

(c) **Multiplication:** for $\boldsymbol{A}_{n,m}$ and $\boldsymbol{B}_{m,p}$, $\boldsymbol{C} = \boldsymbol{AB}$ is $n \times p$ with

$$c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}, \qquad 1 \le i \le n,\ 1 \le j \le p,$$

the inner product of the i-th row of $\boldsymbol{A}$ and the j-th column of $\boldsymbol{B}$; the number of columns of the first matrix must equal the number of rows of the second (3×3 times 3×2 gives 3×2).

Example 1.1. $\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$, $\boldsymbol{B} = \begin{pmatrix} -1 & -2 \\ -3 & -4 \\ -5 & -6 \end{pmatrix}$, $\boldsymbol{C} = \boldsymbol{AB}$. Then $c_{21} = a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} = 4(-1) + 5(-3) + 6(-5)$.

Multiplication is **not commutative**, even for square matrices: $\boldsymbol{AB} \neq \boldsymbol{BA}$.

Example 1.2. $\boldsymbol{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $\boldsymbol{B} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$: $\boldsymbol{AB} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ but $\boldsymbol{BA} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$.

Multiplication is associative: if $\boldsymbol{AB}$ and $\boldsymbol{BC}$ are legal then $(\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{BC})$.

(d) **Scaling:** $(c\boldsymbol{A})_{ij} = c\,a_{ij}$ for a real number c.

(e) **Inversion:** for a square $\boldsymbol{A}$, if there is $\boldsymbol{B}$ of the same shape with $\boldsymbol{AB} = \boldsymbol{I}$ (identity matrix: ones on the diagonal, zeros elsewhere), then $\boldsymbol{B}$ is the inverse $\boldsymbol{A}^{-1}$. If $\boldsymbol{AB} = \boldsymbol{I}$ then $\boldsymbol{BA} = \boldsymbol{I}$, and the inverse is unique: $\boldsymbol{AA}^{-1} = \boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}_n$. With row vectors $\boldsymbol{a}_i = (a_{i1}, \ldots, a_{in})$ of $\boldsymbol{A}_{n,n}$, $\boldsymbol{A}$ is invertible iff the only solution of $\sum_{i=1}^n k_i \boldsymbol{a}_i = \boldsymbol{0}$ is $k_1 = \cdots = k_n = 0$ (*linear independence*). The largest number of independent rows is the *row rank*; row rank = column rank (also for non-square matrices), so one simply speaks of the *rank*. If $\boldsymbol{A}_{n,n}$ is invertible ($\operatorname{rank} \boldsymbol{A} = n$, full rank), then $\boldsymbol{Ax} = \boldsymbol{b}$ has the solution $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$: multiply $\boldsymbol{Ax} = \boldsymbol{b}$ on the left by $\boldsymbol{A}^{-1}$.

1.3 Properties of matrix operations

⟵ **Motivation**
These four identities are used again and again in the regression and GLM derivations.

1. $(\boldsymbol{A} \pm \boldsymbol{B})^\top = \boldsymbol{A}^\top \pm \boldsymbol{B}^\top$.
2. $(\boldsymbol{AB})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$.
3. $(\boldsymbol{AB})^{-1} = \boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$.
4. $(\boldsymbol{A}^\top)^{-1} = (\boldsymbol{A}^{-1})^\top$, often written $\boldsymbol{A}^{-\top}$.

Even if $\boldsymbol{A}$ and $\boldsymbol{B}$ are invertible, $\boldsymbol{A} \pm \boldsymbol{B}$ may not be: $\boldsymbol{A} = \boldsymbol{I}$, $\boldsymbol{B} = -\boldsymbol{I}$.

2 Multivariable calculus

2.1 Quadratic forms

For an $n \times n$ symmetric $\boldsymbol{A}$ and vector $\boldsymbol{x}$,

$$f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{Ax} = (x_1, \ldots, x_n)\,\boldsymbol{A} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \sum_{i,j=1}^n a_{ij} x_i x_j$$

is a *quadratic form*. If for all $\boldsymbol{x} \neq \boldsymbol{0}$:

1. $f(\boldsymbol{x}) > 0$: $\boldsymbol{A}$ is **positive definite**, $\boldsymbol{A} \succ 0$;
2. $f(\boldsymbol{x}) < 0$: **negative definite**, $\boldsymbol{A} \prec 0$;
3. $f(\boldsymbol{x}) \geq 0$: **positive semi-definite**, $\boldsymbol{A} \succeq 0$; similarly $f(\boldsymbol{x}) \leq 0$: negative semi-definite, $\boldsymbol{A} \preceq 0$;
4. otherwise (both signs occur): **indefinite**.

2.2 Polynomials and gradients

⟵ **My thought**
$f(x) = ax + b$ and $f(x) = ax^2 + bx + c$ in one variable; $f(x) = ax^2$ has $f'(x) = 2ax$, which is the symmetric case $2\boldsymbol{Ax}$.

Linear function (degree 1): $P_1(x_1, \ldots, x_n) = \sum_{i=1}^n a_i x_i = \boldsymbol{a}^\top \boldsymbol{x}$. Quadratic (degree 2): $P_2(x_1, \ldots, x_n) = \sum_{i,j} a_{ij} x_i x_j = \boldsymbol{x}^\top \boldsymbol{Ax}$. Gradients:

$$\nabla P_1 = \left(\frac{\partial P_1}{\partial x_1}, \ldots, \frac{\partial P_1}{\partial x_n}\right)^\top = \boldsymbol{a}, \quad \nabla P_2 = \boldsymbol{Ax} + \boldsymbol{A}^\top \boldsymbol{x}\ (= 2\boldsymbol{Ax} \text{ if } \boldsymbol{A} = \boldsymbol{A}^\top).$$

2.3 Taylor expansion

Around a point $\boldsymbol{o}$,

$$\begin{aligned} f(x_1,\dots,x_n) &= f(\boldsymbol{o}) \quad \text{(constant term)} \\ &\quad + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\boldsymbol{o})(x_i - o_i) \quad \text{(linear term)} \\ &\quad + \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}(x_i - o_i)\frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{o})(x_j - o_j) \quad \text{(quadratic term)} \\ &\quad + o\big(\|\boldsymbol{x} - \boldsymbol{o}\|^2\big) \quad \text{(error term)} \\ &= f(\boldsymbol{o}) + \nabla f(\boldsymbol{o})^\top(\boldsymbol{x} - \boldsymbol{o}) + \tfrac{1}{2}(\boldsymbol{x} - \boldsymbol{o})^\top H(f)(\boldsymbol{o})(\boldsymbol{x} - \boldsymbol{o}) + o\big(\|\boldsymbol{x} - \boldsymbol{o}\|^2\big), \end{aligned}$$

where $\nabla f(\boldsymbol{o}) = \big(\frac{\partial f}{\partial x_1}(\boldsymbol{o}), \dots, \frac{\partial f}{\partial x_n}(\boldsymbol{o})\big)^\top$ is the gradient and $H(f)(\boldsymbol{o}) = \big(\frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{o})\big)_{n\times n}$ is the **Hessian matrix**.

2.4 Maximum and minimum points

Sufficient conditions for a strict local extremum at $\boldsymbol{o}$:

1. strict local **maximum**: $\frac{\partial f}{\partial x_i}(\boldsymbol{o}) = 0$ for all i and $\sum_i \sum_j (x_i - o_i)\frac{\partial^2 f}{\partial x_i \partial x_j}(\boldsymbol{o})(x_j - o_j) < 0$ for all $\boldsymbol{x} \neq \boldsymbol{o}$, i.e. $\nabla f(\boldsymbol{o}) = \boldsymbol{0}$ and $H(f)(\boldsymbol{o}) \prec 0$;
2. strict local **minimum**: $\nabla f(\boldsymbol{o}) = \boldsymbol{0}$ and $H(f)(\boldsymbol{o}) \succ 0$.

3 Probability theory and mathematical statistics

3.1 Multivariate normal distribution

For a random vector $\boldsymbol{X} = (X_1, \dots, X_n) \sim \mathrm{MVN}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$,

$$p(x_1,\dots,x_n) = \frac{1}{\sqrt{(2\pi)^n \det(\boldsymbol{\Sigma})}} \exp\Big\{ -\tfrac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu})\Big\}, \qquad \boldsymbol{x} \in \mathbb{R}^n,$$

where $\boldsymbol{\mu} = \mathbb{E}(X_1, \dots, X_n)$ is the mean vector and

$$\boldsymbol{\Sigma} = \begin{pmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots & \mathrm{Cov}(X_1, X_n) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots & \mathrm{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{Cov}(X_n, X_1) & \mathrm{Cov}(X_n, X_2) & \cdots & \mathrm{Var}(X_n) \end{pmatrix}$$

is the covariance matrix. The exponent generalizes $\sum(x-\mu)^2/\sigma^2$. For any random vector the covariance matrix is non-negative (positive semi-) definite.

Proof that $\boldsymbol{\Sigma} = \mathrm{Var}(\boldsymbol{X}) = \mathbb{E}[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top]$ is positive semi-definite. $\boldsymbol{\Sigma} \succeq 0$ iff $\boldsymbol{c}^\top \boldsymbol{\Sigma} \boldsymbol{c} \geq 0$ for all $\boldsymbol{c}$. Now

⟵ **Proof idea**

Turn the quadratic form $c^\top \Sigma c$ into the variance of a scalar random variable.

$$\boldsymbol{c}^\top \boldsymbol{\Sigma} \boldsymbol{c} = \boldsymbol{c}^\top \mathbb{E}[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top]\boldsymbol{c} = \mathbb{E}\big[\underbrace{\boldsymbol{c}^\top(\boldsymbol{X} - \boldsymbol{\mu})}_{Y}\ \underbrace{(\boldsymbol{X} - \boldsymbol{\mu})^\top \boldsymbol{c}}_{Y^\top}\big]$$

$$= \mathbb{E}[YY^\top] = \mathbb{E}(Y^2) \geq 0,$$

where $Y = \boldsymbol{c}^\top(\boldsymbol{X} - \boldsymbol{\mu}) = \sum_{i=1}^n c_i(X_i - \mu_i)$ is a scalar random variable. □

Examples.

1. $X \sim N(\mu, \sigma^2)$: $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\big(-\frac{(x-\mu)^2}{2\sigma^2}\big)$, $x \in \mathbb{R}$.

2. $(X_1, X_2)^\top \sim \mathrm{MVN}_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$: $f_{\boldsymbol{X}}(x_1, x_2) = \frac{1}{2\pi\sqrt{\det \boldsymbol{\Sigma}}} \exp\big(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\big)$.
3. $(X_1, \ldots, X_n)^\top \sim \mathrm{MVN}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$: $f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{(2\pi)^{n/2}(\det \boldsymbol{\Sigma})^{1/2}} \exp\big(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\big)$.
4. **Linear transformation.** If $\boldsymbol{X}$ is an n-dimensional normal vector and $\boldsymbol{A}$ is $m \times n$, then $\boldsymbol{AX} \sim \mathrm{MVN}_m(\boldsymbol{A\mu}, \boldsymbol{A\Sigma A}^\top)$.

Proof of (4). Let $\boldsymbol{Y} = \boldsymbol{AX}$ ($m \times 1 = m \times n \cdot n \times 1$). (i) Mean: $\mathbb{E}(\boldsymbol{Y}) = \mathbb{E}(\boldsymbol{AX}) = \boldsymbol{A}\mathbb{E}(\boldsymbol{X}) = \boldsymbol{A\mu}$, since $\boldsymbol{AX}$ is a linear transformation. (ii) Covariance:

$$\mathrm{Var}(\boldsymbol{Y}) = \mathbb{E}[(\boldsymbol{Y}-\boldsymbol{A\mu})(\boldsymbol{Y}-\boldsymbol{A\mu})^\top] = \mathbb{E}[\boldsymbol{A}(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{A}(\boldsymbol{X}-\boldsymbol{\mu}))^\top]$$

$$= \mathbb{E}[\boldsymbol{A}(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^\top \boldsymbol{A}^\top] = \boldsymbol{A}\,\mathbb{E}[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^\top]\,\boldsymbol{A}^\top = \boldsymbol{A\,\Sigma\,A}^\top.$$

(iii) Normality follows from the linear property of the normal distribution. □

3.2 Sampling distributions: $\chi^2(n)$

1. $Z_1, \ldots, Z_n$ i.i.d. standard normal: $\sum_{i=1}^n Z_i^2 \sim \chi^2(n)$. In matrix form, $\boldsymbol{Z}^\top \boldsymbol{Z} \sim \chi^2(n)$, and $\mathbb{E}[\chi^2(n)] = n$.
2. $\boldsymbol{Y} = (Z_1 + \mu_1, \ldots, Z_n + \mu_n)^\top$ with $Z_i \overset{\text{iid}}{\sim} N(0,1)$: $\boldsymbol{Y}^\top \boldsymbol{Y} = \sum_i (Z_i + \mu_i)^2 \sim \chi^2(n, \lambda)$, the noncentral chi-square with noncentrality parameter $\lambda = \sum_{i=1}^n \mu_i^2$ and $\mathbb{E}[\chi^2(n,\lambda)] = n + \lambda$.
3. $\boldsymbol{X} \sim \mathrm{MVN}_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$: $(\boldsymbol{X}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{X}-\boldsymbol{\mu}) \sim \chi^2(n)$.

Proof of (3). Write $(\boldsymbol{X}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{X}-\boldsymbol{\mu}) = \underbrace{(\boldsymbol{X}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1/2}}_{\boldsymbol{Z}^\top} \underbrace{\boldsymbol{\Sigma}^{-1/2}(\boldsymbol{X}-\boldsymbol{\mu})}_{\boldsymbol{Z}}$ and denote $\boldsymbol{Z} = \boldsymbol{\Sigma}^{-1/2}(\boldsymbol{X}-\boldsymbol{\mu})$. Then

← **Proof idea**
Split $\Sigma^{-1} = \Sigma^{-1/2}\Sigma^{-1/2}$ and standardize.

$$\mathbb{E}(\boldsymbol{Z}) = \boldsymbol{\Sigma}^{-1/2}\big(\mathbb{E}(\boldsymbol{X}) - \boldsymbol{\mu}\big) = \boldsymbol{0},$$

$$\mathrm{Var}(\boldsymbol{Z}) = \boldsymbol{\Sigma}^{-1/2}\,\mathrm{Var}(\boldsymbol{X})(\boldsymbol{\Sigma}^{-1/2})^\top = \boldsymbol{\Sigma}^{-1/2}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-1/2} = \boldsymbol{I},$$

so $\boldsymbol{Z} \sim \mathrm{MVN}(\boldsymbol{0}, \boldsymbol{I}_n)$ and $(\boldsymbol{X}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{X}-\boldsymbol{\mu}) = \boldsymbol{Z}^\top \boldsymbol{Z} = \sum_{i=1}^n Z_i^2 \sim \chi^2(n)$. □

3.3 t-distribution and F-distribution

1. Independent $X \sim N(0,1)$ and $Y \sim \chi^2(n)$: $T = \dfrac{X}{\sqrt{Y/n}} \sim t(n)$, Student's t with n degrees of freedom; symmetric about zero; its density converges to $N(0,1)$ as $n \to \infty$.
2. Independent $U \sim \chi^2(n)$ and $V \sim \chi^2(m)$: $\dfrac{U/n}{V/m} \sim F(n,m)$.
3. Additive property of χ^2: independent $U \sim \chi^2(n)$, $V \sim \chi^2(m)$ give $U + V \sim \chi^2(n+m)$.

3.4 Parameter estimation

For independent samples $X_1, \ldots, X_n$ of a distribution with parameter θ, an estimator is $\hat{\theta} = \hat{\theta}_n = \hat{\theta}_n(X_1, \ldots, X_n) = \hat{\theta}_n(\boldsymbol{X})$.

1. Estimation error: $\tilde{\theta}_n = \hat{\theta}_n - \theta$.
2. Bias: $b_\theta(\hat{\theta}_n) = \mathbb{E}_\theta[\hat{\theta}_n] - \theta$.
3. Unbiased if $\mathbb{E}_\theta[\hat{\theta}_n] = \theta$; asymptotically unbiased if $\lim_{n\to\infty} \mathbb{E}_\theta[\hat{\theta}_n] = \theta$.
4. Mean squared error $\mathrm{MSE}_\theta(\hat{\theta}_n) = \mathbb{E}_\theta[(\hat{\theta}_n - \theta)^2]$:

$$\mathbb{E}_\theta[(\tilde{\theta}_n)^2] \;=\; (\mathbb{E}_\theta[\tilde{\theta}_n])^2 \;+\; \mathrm{Var}_\theta(\hat{\theta}_n) \;=\; (b_\theta(\hat{\theta}_n))^2 \;+\; \mathrm{Var}_\theta(\hat{\theta}_n)$$

Hence there is usually a trade-off between the bias and the variance of $\hat{\theta}$.

3.5 Confidence interval

Fix a confidence level $1-\alpha$ (α typically $0.1, 0.05, 0.01$). If two estimators $\hat{\theta}_n^-(\boldsymbol{X}) \le \hat{\theta}_n^+(\boldsymbol{X})$ satisfy $\mathbb{P}_\theta(\hat{\theta}_n^- \le \theta \le \hat{\theta}_n^+) \ge 1-\alpha$, then $[\hat{\theta}_n^-, \hat{\theta}_n^+]$ is a $1-\alpha$ confidence interval. To find one:

1. construct a random function $U(X_1, \dots, X_n, \theta)$ whose only unknown is θ and whose distribution is known (a *pivotal quantity*);
2. find constants a, b with $\mathbb{P}(a < U < b) = 1-\alpha$ and solve $a < U < b$ for $\hat{\theta}_n^-$, $\hat{\theta}_n^+$.

Remark. This does not mean $\theta \in [\hat{\theta}_n^-, \hat{\theta}_n^+]$ with probability $\ge 1-\alpha$ for the given sample: for given samples the endpoints are deterministic. Roughly, with samples $\boldsymbol{X}_1, \dots, \boldsymbol{X}_m$, about $(1-\alpha)m$ of the intervals $[\hat{\theta}_n^-(\boldsymbol{X}_i), \hat{\theta}_n^+(\boldsymbol{X}_i)]$ contain θ.

3.6 Hypothesis testing

Two-tailed test: $H_0 : \theta = \theta_0$ (null) against $H_1 : \theta \ne \theta_0$ (alternative), at significance level α:

1. under H_0 construct a statistic U with known distribution (a pivotal quantity);
2. find a threshold k with $\mathbb{P}_{\theta_0}(|U| \ge k) \le \alpha$; compute $U(\boldsymbol{X})$; reject H_0 if $U(\boldsymbol{X})$ falls in the rejection region $\{|U| \ge k\}$.

Equivalently, the p-value approach: $p = \mathbb{P}_{\theta_0}(|U| \ge U(\boldsymbol{X}))$; reject H_0 if $p \le \alpha$ ($p \le \alpha \Leftrightarrow U(\boldsymbol{X}) \ge k$).

Part II. Linear regression models

4 From a mean to a regression model

Throughout, n is the number of observations and p counts slopes, so a design matrix with an intercept has $k = p + 1$ columns.

⟵ **Proof idea**
Check the initial and boundary conditions after finding a general solution.

A sample mean predicts every observation in the same way; it is useful when observations can be treated as exchangeable around one common mean. If a measured covariate explains systematic differences, a *conditional mean* is more informative,

$$m(x) = E(Y \mid X = x) = \int y\, h(y \mid x)\, dy,$$

and, as a comparison, the conditional median $m(x)$ solves $\int_{-\infty}^{m(x)} h(y \mid x)\, dy = \frac{1}{2}$. For many points there are many possible regression functions f (nonparametric), so it is useful to restrict to simple functions described by a few parameters. True regression functions are never exactly linear, but we assume the dependence of Y on $X_1, \ldots, X_p$ is linear. In simple linear regression (SLR) the line is a model for the *mean*, while the random error describes variation around it:

⟵ **My thought**
Decomposing a complicated signal into simpler modes makes dynamics easier to understand.

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad E(\varepsilon_i \mid X) = 0, \qquad \mathrm{Var}(\varepsilon_i \mid X) = \sigma^2.$$

For classical normal theory, condition on the design and assume $\varepsilon \sim N_n(0, \sigma^2 I_n)$. Least squares itself does not require normality; the exact small-sample t and F distributions do. A slope is the change in the modeled mean per unit change in x; with several predictors its interpretation is conditional on the others being held fixed. An intercept is the modeled mean when all included predictors are zero, which need not be scientifically meaningful.

4.1 Least squares, fitted values and residuals

The least-squares line minimizes $\mathrm{RSS}(b_0, b_1) = \sum_i (y_i - b_0 - b_1 x_i)^2$. With $S_{xx} = \sum_i (x_i - \bar{x})^2$ and $S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y})$,

⟵ **Motivation**
Why minimize distance? Orthogonality turns a nonlinear-looking optimization into normal equations.

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \qquad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \qquad e_i = y_i - \hat{y}_i.$$

The fitted line passes through $(\bar{x}, \bar{y})$. Residuals are observed differences from the fitted line; they are not the unobserved errors ε_i. With an intercept the normal equations give $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$. Geometrically $\hat{y}$ is the orthogonal projection of y onto the column space of the design matrix and e is perpendicular to that space.

OLS and its history. Ordinary least squares was proposed by Legendre (1805) and developed by Gauss (1809), before the systematic understanding of conditional expectation. Afterwards it was recognized that the OLS regression equation estimates $E(Y \mid X = x)$:

- **OLS estimate:** $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$, based on the smallest RSS.
- **Conditional expectation:** under the model, $E(Y \mid X = x) = \beta_0 + \beta_1 x$ is the *real regression line*, which cannot be seen in practice; $\hat{Y}$ is its estimate.

Under the classical assumptions OLS (which coincides with MLE in normal SLR) gives unbiased estimators of the real regression line based on $E(y_i \mid x_i)$.

Analogy: the fitted line plays the role of a sample mean for a population mean in a sampling distribution.

Essence of SLR through the correlation coefficient ρ. Standardizing both variables,

$$\frac{\hat{y}-\bar{y}}{sd(y)} = \rho\,\frac{x-\bar{x}}{sd(x)},$$

so the slope on the standardized scale is the correlation; the proof uses covariance in the bivariate normal case (Teacher: Shisong Mao).

4.2 Variation and the ANOVA comparison in SLR

With an intercept, $y_i-\bar{y} = (\hat{y}_i-\bar{y}) + e_i$ and the two terms are orthogonal, so

$$\underbrace{\sum_i (y_i-\bar{y})^2}_{\text{TSS}} = \underbrace{\sum_i (\hat{y}_i-\bar{y})^2}_{\text{RegSS}} + \underbrace{\sum_i e_i^2}_{\text{RSS}}, \qquad R^2 = 1-\frac{\text{RSS}}{\text{TSS}}.$$

TSS has $n-1$ degrees of freedom. For SLR, RegSS has 1 and RSS has $n-2$; for full-rank multiple regression with p slopes these become p and $n-p-1$. The residual mean square $s^2 = \text{RSS}/(n-p-1)$ estimates σ^2. The global statistic $F = \dfrac{\text{RegSS}/p}{\text{RSS}/(n-p-1)}$ tests whether all slopes are zero. R^2 is the fraction of sample variation explained relative to the intercept-only baseline; it does not by itself establish causality or adequacy of assumptions. In SLR the two-sided slope t test and the global F test agree because $F = t^2$.

4.3 Uncertainty for a coefficient and for a future response

Under normal SLR assumptions, $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$, so a t statistic replaces σ with s; a two-sided confidence interval is $\hat{\beta}_1 \pm t_{n-2,1-\alpha/2}\, s/\sqrt{S_{xx}}$. At x_*, $\hat{m}(x_*) = \hat{\beta}_0 + \hat{\beta}_1 x_*$ has estimated standard error $s\sqrt{1/n + (x_*-\bar{x})^2/S_{xx}}$, whereas the predictive standard error for a future Y_* is

$$s\sqrt{1+\frac{1}{n}+\frac{(x_*-\bar{x})^2}{S_{xx}}}.$$

A prediction interval uses $\hat{m}(x_*) \pm t_{n-2,1-\alpha/2}$ times this second standard error; it is wider because it includes the future observation's intrinsic variation.

← **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

4.4 Hypothesis tests: z, t, F and the noncentrality parameter

Slope, $H_0 : \beta_1 = 0$. Since $y_i \sim \text{NID}(\beta_0+\beta_1 x_i, \sigma^2)$, $\hat{\beta}_1 \sim N\big(\beta_1, \sigma^2/\sum(x_i-\bar{x})^2\big)$. If σ is known use $z = \hat{\beta}_1/\sqrt{\sigma^2/\sum(x_i-\bar{x})^2}$; otherwise

$$t_0 = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)},$$

substituting the unbiased estimate $\sum(y_i-\hat{y}_i)^2/(n-2)$ for σ^2.

← **Motivation**
From a probabilistic model to a usable numerical quantity: what is observed, what is random?

F of the analysis of variance. Two readings: model interpretation (explained versus unexplained variance) and a hypothesis test with noncentrality parameter zero or not:

$$F = \frac{\sum(\hat{y}_i-\bar{y})^2/(\sigma^2 p)}{\sum(y_i-\hat{y}_i)^2/(\sigma^2(n-p-1))} \sim F_{p,\,n-p-1} \text{ under } H_0.$$

If H_0 is false, F follows a noncentral F with 1 and $n-2$ degrees of freedom in SLR, with noncentrality parameter

$$\lambda = \beta_1^2 \frac{\sum (x_i - \bar{x})^2}{\sigma^2} \neq 0.$$

Since F measures the model's explanatory ability, $F > 1$ (or above a threshold) indicates a useful model. For a single variable the t test is more adaptable because it allows one-sided alternatives, although it is equivalent to F in SLR.

5 Multiple linear regression: model and least squares

5.1 Notation and the model

Regression geometry

I understand regression by projecting y onto the columns of X. The residual is the orthogonal component.

With p explanatory variables and n independent observations,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where $\varepsilon_1, \ldots, \varepsilon_n$ are independent, $\mathbb{E}[\varepsilon_i] = 0$, $\mathrm{Var}(\varepsilon_i) = \sigma^2$, and for inference $\varepsilon_i \sim N(0, \sigma^2)$. The regression function is $\mathbb{E}[y_i \mid x_{i1}, \ldots, x_{ip}] = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$; β_j is the change in the mean response for a one-unit increase in x_j *holding all other predictors fixed*. Stacking the n equations,

$$\underbrace{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}}_{\boldsymbol{y}\ (n\times 1)} = \underbrace{\begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}}_{\boldsymbol{X}\ (n\times(p+1))} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}}_{\boldsymbol{\beta}\ ((p+1)\times 1)} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}}_{\boldsymbol{\varepsilon}\ (n\times 1)}.$$

Write $k := p + 1$ for the number of regression parameters. Compactly,

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \mathbb{E}[\boldsymbol{\varepsilon}] = \boldsymbol{0}, \qquad \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{I}_n, \qquad \boldsymbol{\varepsilon} \sim N_n(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_n) \text{ for inference.}$$

$\boldsymbol{X}$ is the *design matrix*; it always contains a column of ones below (the model has an intercept).

Remark 5.1 (Full rank). All algebra below assumes $\mathrm{rank}(\boldsymbol{X}) = k = p + 1$: the columns of $\boldsymbol{X}$ are linearly independent, which is exactly the condition for $\boldsymbol{X}^\top \boldsymbol{X}$ to be invertible. It fails if one predictor is a linear combination of the others, or if a categorical predictor with r levels is coded with r dummies while an intercept is present (the "dummy-variable trap").

5.2 The least squares criterion and the normal equations

Definition 5.2 (Least squares estimator)**.** $\hat{\boldsymbol{\beta}}$ minimises the residual sum of squares

$$SS(\boldsymbol{\beta}) = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})^\top (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big(y_i - \sum_{j=0}^{p} \beta_j x_{ij} \Big)^2, \qquad x_{i0} := 1.$$

Theorem 5.3 (Normal equations)**.** *If* $\mathrm{rank}(\boldsymbol{X}) = p + 1$ *then* $SS(\boldsymbol{\beta})$ *has the unique minimiser*

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$$

and $\hat{\boldsymbol{\beta}}$ *solves the normal equations* $\boldsymbol{X}^\top \boldsymbol{X} \boldsymbol{\beta} = \boldsymbol{X}^\top \boldsymbol{y}$.

Proof. Expand: $SS(\boldsymbol{\beta}) = \boldsymbol{y}^\top\boldsymbol{y} - 2\boldsymbol{\beta}^\top\boldsymbol{X}^\top\boldsymbol{y} + \boldsymbol{\beta}^\top\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\beta}$. Two matrix-calculus facts, obtained by differentiating componentwise:

$$\frac{\partial(\boldsymbol{a}^\top\boldsymbol{\beta})}{\partial\boldsymbol{\beta}} = \boldsymbol{a}, \qquad \frac{\partial(\boldsymbol{\beta}^\top\boldsymbol{A}\boldsymbol{\beta})}{\partial\boldsymbol{\beta}} = (\boldsymbol{A} + \boldsymbol{A}^\top)\boldsymbol{\beta}.$$

(For the second: $\boldsymbol{\beta}^\top\boldsymbol{A}\boldsymbol{\beta} = \sum_{r,s} A_{rs}\beta_r\beta_s$ and $\partial/\partial\beta_j$ of that is $\sum_s A_{js}\beta_s + \sum_r A_{rj}\beta_r$, the j-th entry of $(\boldsymbol{A} + \boldsymbol{A}^\top)\boldsymbol{\beta}$.) With $\boldsymbol{A} = \boldsymbol{X}^\top\boldsymbol{X}$ symmetric,

$$\frac{\partial SS(\boldsymbol{\beta})}{\partial\boldsymbol{\beta}} = -2\boldsymbol{X}^\top\boldsymbol{y} + 2\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\beta} = -2\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}).$$

Setting this to zero gives $\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}^\top\boldsymbol{y}$. Because $\operatorname{rank}(\boldsymbol{X}) = p + 1$, $\boldsymbol{X}^\top\boldsymbol{X}$ is positive definite and invertible, so the unique stationary point is $\hat{\boldsymbol{\beta}}$. The Hessian $\partial^2 SS/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top = 2\boldsymbol{X}^\top\boldsymbol{X}$ is positive definite, so it is a minimum (the unique global minimum: SS is a strictly convex quadratic). □

Remark 5.4 (Component form of the first-order conditions). Row by row the normal equations read

$$\sum_i e_i = 0, \qquad \sum_i x_{i1}e_i = 0, \quad \ldots, \quad \sum_i x_{ip}e_i = 0, \qquad e_i := y_i - \hat{y}_i,\ \hat{y}_i = \boldsymbol{x}_i^\top\hat{\boldsymbol{\beta}},$$

i.e. the single statement $\boldsymbol{X}^\top\boldsymbol{e} = \boldsymbol{0}$. This is the reason the ANOVA decomposition works.

5.3 Fitted values, residuals and the hat matrix

Definition 5.5. $\hat{\boldsymbol{y}} := \boldsymbol{X}\hat{\boldsymbol{\beta}}$, $\boldsymbol{e} := \boldsymbol{y} - \hat{\boldsymbol{y}}$, $\boldsymbol{H} := \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top$. $\boldsymbol{H}$ is the *hat matrix* because $\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y}$: it puts the hat on $\boldsymbol{y}$.

Proposition 5.6 (Properties of $\boldsymbol{H}$)**.** *$\boldsymbol{H}$ is symmetric and idempotent: $\boldsymbol{H}^\top = \boldsymbol{H}$, $\boldsymbol{H}^2 = \boldsymbol{H}$. Consequently $\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y}$, $\boldsymbol{e} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}$, $\boldsymbol{I}_n - \boldsymbol{H}$ is also symmetric and idempotent, $\boldsymbol{H}(\boldsymbol{I}_n - \boldsymbol{H}) = \boldsymbol{0}$, and $\operatorname{tr}(\boldsymbol{H}) = \operatorname{rank}(\boldsymbol{H}) = p + 1 = k$, hence $\operatorname{tr}(\boldsymbol{I}_n - \boldsymbol{H}) = n - k$.*

Proof. Symmetry: $\boldsymbol{H}^\top = (\boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top)^\top = \boldsymbol{X}((\boldsymbol{X}^\top\boldsymbol{X})^{-1})^\top\boldsymbol{X}^\top = \boldsymbol{H}$ because $\boldsymbol{X}^\top\boldsymbol{X}$ and its inverse are symmetric. Idempotence:

$$\boldsymbol{H}^2 = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\underbrace{\boldsymbol{X}^\top\boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}}_{\boldsymbol{I}}\boldsymbol{X}^\top = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top = \boldsymbol{H}.$$

Then $\boldsymbol{e} = \boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}$ and $(\boldsymbol{I}_n - \boldsymbol{H})^2 = \boldsymbol{I}_n - 2\boldsymbol{H} + \boldsymbol{H}^2 = \boldsymbol{I}_n - \boldsymbol{H}$. For any idempotent matrix $\operatorname{tr} = \operatorname{rank}$, and $\operatorname{rank}(\boldsymbol{H}) = \operatorname{rank}(\boldsymbol{X}) = p + 1$ because $\boldsymbol{H}$ is the orthogonal projector onto the column space of $\boldsymbol{X}$. Hence $\operatorname{tr}(\boldsymbol{I}_n - \boldsymbol{H}) = n - k$. □

Proposition 5.7 (Two consequences of the normal equations)**.** *With $\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}}$: $\boldsymbol{X}^\top\boldsymbol{e} = \boldsymbol{0}$, so $\sum_{i=1}^n e_i = 0$ and $\sum_{i=1}^n \hat{y}_i e_i = 0$; and*

$$\hat{\boldsymbol{y}}^\top\hat{\boldsymbol{y}} + \boldsymbol{e}^\top\boldsymbol{e} = \boldsymbol{y}^\top\boldsymbol{y}, \qquad \textit{i.e.} \qquad \sum_i \hat{y}_i^2 + \sum_i e_i^2 = \sum_i y_i^2.$$

Proof. $\boldsymbol{X}^\top\boldsymbol{e} = \boldsymbol{X}^\top\boldsymbol{y} - \boldsymbol{X}^\top\boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{0}$ by the normal equations. The first column of $\boldsymbol{X}$ is $\boldsymbol{1}$, so the first entry of $\boldsymbol{X}^\top\boldsymbol{e}$ is $\sum_i e_i = 0$; and $\sum_i \hat{y}_i e_i = \hat{\boldsymbol{y}}^\top\boldsymbol{e} = \hat{\boldsymbol{\beta}}^\top\boldsymbol{X}^\top\boldsymbol{e} = 0$. Since $\boldsymbol{y} = \hat{\boldsymbol{y}} + \boldsymbol{e}$ with orthogonal pieces, $\boldsymbol{y}^\top\boldsymbol{y} = \hat{\boldsymbol{y}}^\top\hat{\boldsymbol{y}} + 2\hat{\boldsymbol{y}}^\top\boldsymbol{e} + \boldsymbol{e}^\top\boldsymbol{e} = \hat{\boldsymbol{y}}^\top\hat{\boldsymbol{y}} + \boldsymbol{e}^\top\boldsymbol{e}$. □

5.4 Mean and covariance of the least squares estimator

Theorem 5.8 (Unbiasedness and covariance)**.** *Under $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbb{E}[\boldsymbol{\varepsilon}] = \boldsymbol{0}$, $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{I}_n$:*

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}, \qquad \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1}.$$

If moreover $\boldsymbol{\varepsilon} \sim N_n(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_n)$, then $\hat{\boldsymbol{\beta}} \sim N_{p+1}\big(\boldsymbol{\beta}, \sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\big)$.

Proof. $\hat{\boldsymbol{\beta}} = \boldsymbol{A}\boldsymbol{y}$ with $\boldsymbol{A} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top$ and $\boldsymbol{A}\boldsymbol{X} = \boldsymbol{I}$. Hence $\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{A}\mathbb{E}[\boldsymbol{y}] = \boldsymbol{A}\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{\beta}$. Using $\mathrm{Var}(\boldsymbol{A}\boldsymbol{y}) = \boldsymbol{A}\,\mathrm{Var}(\boldsymbol{y})\boldsymbol{A}^\top$,

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \boldsymbol{A}\, \sigma^2 \boldsymbol{I}_n\, \boldsymbol{A}^\top = \sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{X} (\boldsymbol{X}^\top \boldsymbol{X})^{-1} = \sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1}.$$

Finally $\hat{\boldsymbol{\beta}}$ is a fixed linear transformation of the normal vector $\boldsymbol{y}$, so it is normal with this mean and covariance. □

Remark 5.9 (Reading the covariance matrix)*.* The $(j+1)$-th diagonal element of $\sigma^2 (\boldsymbol{X}^\top \boldsymbol{X})^{-1}$ is $\mathrm{Var}(\hat{\beta}_j)$, so

$$\widehat{\mathrm{se}}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(\boldsymbol{X}^\top \boldsymbol{X})^{-1}]_{jj}} = \sqrt{\mathrm{MSE}\,[(\boldsymbol{X}^\top \boldsymbol{X})^{-1}]_{jj}},$$

with MSE justified in Section 6. Off-diagonal elements are covariances $\mathrm{Cov}(\hat{\beta}_j, \hat{\beta}_l) = \sigma^2 [(\boldsymbol{X}^\top \boldsymbol{X})^{-1}]_{jl}$.

Interactions. A term $x_1 x_2$ lets the slope of one continuous predictor depend on the other. With a binary z, $\beta_0 + \beta_1 x + \beta_2 z + \beta_3 xz$ gives an intercept change β_2 and a slope change β_3 between groups.

6 The ANOVA table: derivations

Least squares & tests

I separate algebraic ANOVA identities from the Gaussian assumptions behind F and t distributions.

Four logically separate steps: (1) **algebra**: $\mathrm{TSS} = \mathrm{RegSS} + \mathrm{RSS}$ holds for the least squares fit regardless of distribution; (2) **unbiasedness**: $\mathbb{E}[\mathrm{MSE}] = \sigma^2$ needs only the second-moment assumptions, not normality; (3) **distributions**: under normality $\mathrm{RSS}/\sigma^2 \sim \chi^2_{n-k}$, $\mathrm{RegSS}/\sigma^2 \sim \chi^2_p$ under H_0, and the two are independent; (4) **the test**: the ratio has an F distribution.

6.1 Step 1: the ANOVA identity

Definition 6.1 (The three sums of squares)**.**

$$\mathrm{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_i y_i^2 - n\bar{y}^2 = \boldsymbol{y}^\top \boldsymbol{y} - \tfrac{1}{n} \boldsymbol{y}^\top \boldsymbol{1}\boldsymbol{1}^\top \boldsymbol{y},$$

$$\mathrm{RegSS} = \sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2 = \sum_i \hat{y}_i^2 - n\bar{y}^2,$$

$$\mathrm{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \sum_i e_i^2 = \boldsymbol{e}^\top \boldsymbol{e}.$$

TSS measures variation of the response about its mean *before* regression; RSS is what is left *after* regression; RegSS is the part the regression has accounted for.

Theorem 6.2 (ANOVA decomposition)**.** *For the least squares fit of a model with an intercept,* $\mathrm{TSS} = \mathrm{RegSS} + \mathrm{RSS}$*, all three non-negative.*

Proof. Telescoping identity, valid for every i: $y_i - \bar{y} = \underbrace{(y_i - \hat{y}_i)}_{e_i} + (\hat{y}_i - \bar{y})$. Note $\bar{\hat{y}} = \bar{y}$, because $\sum_i \hat{y}_i = \mathbf{1}^\top \boldsymbol{H}\boldsymbol{y}$ and $\mathbf{1}$ is a column of $\boldsymbol{X}$, so $\boldsymbol{H}\mathbf{1} = \mathbf{1}$ and $\sum_i \hat{y}_i = \mathbf{1}^\top \boldsymbol{y} = \sum_i y_i$. Squaring and summing,

$$\mathrm{TSS} = \sum_i e_i^2 + \sum_i (\hat{y}_i - \bar{y})^2 + 2\sum_i e_i(\hat{y}_i - \bar{y}),$$

and the cross term vanishes,

$$\sum_i e_i(\hat{y}_i - \bar{y}) = \sum_i e_i \hat{y}_i - \bar{y}\sum_i e_i = 0 - 0 = 0,$$

by Proposition 5.7. Hence $\mathrm{TSS} = \mathrm{RSS} + \mathrm{RegSS}$. □

Remark 6.3 (Geometric content). $\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y}$ is the projection onto $\mathrm{col}(\boldsymbol{X})$ (the "explained" part), $\boldsymbol{e} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}$ the projection onto its orthogonal complement ("unexplained"). $\mathrm{TSS} = \mathrm{RegSS} + \mathrm{RSS}$ is Pythagoras in $\mathbb{R}^n$ after centring; $\boldsymbol{X}^\top \boldsymbol{e} = \mathbf{0}$ and $\sum_i e_i = 0$ are the only two facts needed.

Remark 6.4 (Two useful alternative forms). Using $\boldsymbol{X}^\top \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}^\top \boldsymbol{y}$,

$$\mathrm{RegSS} = \hat{\boldsymbol{\beta}}^\top \boldsymbol{X}^\top \boldsymbol{y} - n\bar{y}^2 = \hat{\boldsymbol{\beta}}^\top \boldsymbol{X}^\top \boldsymbol{y} - \tfrac{1}{n}\boldsymbol{y}^\top \mathbf{1}\mathbf{1}^\top \boldsymbol{y}, \qquad \mathrm{RegSS} = \hat{\boldsymbol{y}}^\top \hat{\boldsymbol{y}} - \tfrac{1}{n}\boldsymbol{y}^\top \mathbf{1}\mathbf{1}^\top \boldsymbol{y},$$

the second from $\hat{\boldsymbol{y}}^\top \hat{\boldsymbol{y}} = \boldsymbol{y}^\top \boldsymbol{y} - \mathrm{RSS}$. The worked example checks that they agree.

6.2 Step 2: MSE is an unbiased estimator of σ^2

Lemma 6.5 (Expected residual sum of squares)**.** *Under* $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$, $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{I}_n$*:* $\mathbb{E}[\mathrm{RSS}] = \mathbb{E}[\boldsymbol{e}^\top \boldsymbol{e}] = \sigma^2(n - p - 1) = \sigma^2(n - k)$.

Proof. $\boldsymbol{e} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{\varepsilon}$ because $(\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{X}\boldsymbol{\beta} = \mathbf{0}$, so $\boldsymbol{e}^\top \boldsymbol{e} = \boldsymbol{\varepsilon}^\top (\boldsymbol{I}_n - \boldsymbol{H})^\top (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^\top (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{\varepsilon}$. For a quadratic form in a random vector with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$,

$$\boxed{\mathbb{E}[\boldsymbol{\varepsilon}^\top \boldsymbol{M}\boldsymbol{\varepsilon}] \quad = \quad \mathrm{tr}(\boldsymbol{M}\boldsymbol{\Sigma}) \;+\; \boldsymbol{\mu}^\top \boldsymbol{M}\boldsymbol{\mu}}$$

(write $\boldsymbol{\varepsilon}^\top \boldsymbol{M}\boldsymbol{\varepsilon} = \sum_{r,s} M_{rs}\varepsilon_r\varepsilon_s$ and use $\mathbb{E}[\varepsilon_r\varepsilon_s] = \Sigma_{rs} + \mu_r\mu_s$). With $\boldsymbol{M} = \boldsymbol{I}_n - \boldsymbol{H}$, $\boldsymbol{\Sigma} = \sigma^2 \boldsymbol{I}_n$, $\boldsymbol{\mu} = \mathbf{0}$: $\mathbb{E}[\boldsymbol{e}^\top \boldsymbol{e}] = \sigma^2 \,\mathrm{tr}(\boldsymbol{I}_n - \boldsymbol{H}) = \sigma^2(n - \mathrm{tr}\,\boldsymbol{H}) = \sigma^2(n - p - 1)$. □

Corollary 6.6. *Define* $\mathrm{MSE} := \mathrm{RSS}/(n - p - 1)$, $\mathrm{MSR} := \mathrm{RegSS}/p$, $s^2 := \mathrm{MSE}$. *Then* $\mathbb{E}[\mathrm{MSE}] = \sigma^2$*;* $s = \sqrt{\mathrm{MSE}}$ *is the "residual standard error" printed by* `summary()` *in R. Under* $H_0 : \beta_1 = \cdots = \beta_p = 0$, $\mathbb{E}[\mathrm{MSR}] = \sigma^2$.

Proof. The first claim is Lemma 6.5 divided by $n - p - 1$. Under H_0 the regression function is the constant β_0 and $\mathbb{E}[\mathrm{TSS}] = (n-1)\sigma^2$, so $\mathbb{E}[\mathrm{RegSS}] = \mathbb{E}[\mathrm{TSS}] - \mathbb{E}[\mathrm{RSS}] = (n-1)\sigma^2 - (n-p-1)\sigma^2 = p\sigma^2$, hence $\mathbb{E}[\mathrm{MSR}] = \sigma^2$. □

Remark 6.7 (Why the degrees of freedom are what they are). TSS has $n-1$ df (n observations minus one for $\bar{y}$); RSS has $n-p-1$ df (n minus $p+1$ estimated parameters; algebraically $\mathrm{rank}(\boldsymbol{I}_n - \boldsymbol{H}) = n-p-1$); RegSS has p df (the leftover $(n-1) - (n-p-1) = p$, the number of slopes tested; the centred fitted vector lives in a p-dimensional space). Degrees of freedom add: $p + (n-p-1) = n-1$.

6.3 Step 3: distributions of RSS and RegSS, and independence

Now normality enters, through Cochran's theorem in an easily verified form.

Lemma 6.8 (Distribution of a quadratic form)**.** *Let $\boldsymbol{z} \sim N_n(\boldsymbol{0}, \boldsymbol{I}_n)$ and $\boldsymbol{A}$ symmetric idempotent with $\operatorname{rank}(\boldsymbol{A}) = r$. Then $\boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z} \sim \chi^2_r$. If $\boldsymbol{B}$ is symmetric idempotent with $\boldsymbol{A}\boldsymbol{B} = \boldsymbol{0}$, then $\boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z}$ and $\boldsymbol{z}^\top \boldsymbol{B}\boldsymbol{z}$ are independent.*

Proof sketch. Spectral decomposition $\boldsymbol{A} = \boldsymbol{P}\boldsymbol{\Lambda}\boldsymbol{P}^\top$, $\boldsymbol{P}$ orthogonal, $\boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Idempotence forces $\lambda^2 = \lambda$, so $\lambda_j \in \{0, 1\}$ and $\operatorname{rank}(\boldsymbol{A}) = \sum_j \lambda_j = r$. Then

$$\boldsymbol{z}^\top \boldsymbol{A}\boldsymbol{z} = \sum_{j=1}^{n} \lambda_j (\boldsymbol{P}^\top \boldsymbol{z})_j^2 = \sum_{j:\lambda_j = 1} w_j^2, \qquad \boldsymbol{w} = \boldsymbol{P}^\top \boldsymbol{z} \sim N_n(\boldsymbol{0}, \boldsymbol{I}_n),$$

a sum of r independent squared standard normals. If $\boldsymbol{A}\boldsymbol{B} = \boldsymbol{0}$, the columns of $\boldsymbol{P}$ with unit eigenvalues of $\boldsymbol{A}$ are orthogonal to those with unit eigenvalues of $\boldsymbol{B}$, so the two sums use disjoint groups of the independent w_j. □

Theorem 6.9 (Distributions behind the ANOVA table)**.** *Assume $\boldsymbol{\varepsilon} \sim N_n(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_n)$ and $\operatorname{rank}(\boldsymbol{X}) = k$. Then*

$$\frac{\mathrm{RSS}}{\sigma^2} \sim \chi^2_{n-p-1}, \qquad \mathrm{RSS} \text{ is independent of } \hat{\boldsymbol{\beta}} \text{ (hence of } \hat{\boldsymbol{y}} \text{ and } \mathrm{RegSS}).$$

Under $H_0 : \beta_1 = \cdots = \beta_p = 0$: $\mathrm{RegSS}/\sigma^2 \sim \chi^2_p$ and $\mathrm{RegSS} \perp\!\!\!\perp \mathrm{RSS}$.

Proof. (a) *Error line.* Put $\boldsymbol{z} = \boldsymbol{\varepsilon}/\sigma$. Because $(\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{X} = \boldsymbol{0}$, $\boldsymbol{e} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{\varepsilon} = \sigma(\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{z}$; $\boldsymbol{I}_n - \boldsymbol{H}$ is symmetric idempotent of rank $n - k$, so $\mathrm{RSS}/\sigma^2 = \boldsymbol{z}^\top(\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{z} \sim \chi^2_{n-k}$.

(b) *Independence of $\boldsymbol{e}$ and $\hat{\boldsymbol{\beta}}$.* Both are linear in $\boldsymbol{y}$, so jointly normal; zero covariance then implies independence. With $\boldsymbol{A} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top$,

$$\operatorname{Cov}\big((\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}, \boldsymbol{A}\boldsymbol{y}\big) = \sigma^2 (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{A}^\top,$$

$$(\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{A}^\top = \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1} - \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1} = \boldsymbol{0}.$$

Hence RSS is independent of $\hat{\boldsymbol{\beta}}$, so of $\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}}$ and of $\mathrm{RegSS} = \|\hat{\boldsymbol{y}} - \bar{y}\boldsymbol{1}\|^2$.

(c) *Regression line under H_0.* The true model is $\boldsymbol{y} = \boldsymbol{\mu} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\mu} = \beta_0 \boldsymbol{1}$. The centred fitted vector is $\hat{\boldsymbol{y}} - \bar{y}\boldsymbol{1} = (\boldsymbol{H} - \boldsymbol{P}_0)\boldsymbol{y}$ with $\boldsymbol{P}_0 := \frac{1}{n}\boldsymbol{1}\boldsymbol{1}^\top$, the hat matrix of the intercept-only model (symmetric, idempotent, rank 1, and $\boldsymbol{P}_0\boldsymbol{H} = \boldsymbol{P}_0$ because $\boldsymbol{1} \in \operatorname{col}(\boldsymbol{X})$). Put $\boldsymbol{M} := \boldsymbol{H} - \boldsymbol{P}_0$. Then $\boldsymbol{M}$ is symmetric and idempotent, $\boldsymbol{M}^2 = \boldsymbol{H} - \boldsymbol{P}_0 - \boldsymbol{P}_0 + \boldsymbol{P}_0 = \boldsymbol{M}$; $\operatorname{rank}(\boldsymbol{M}) = \operatorname{tr}(\boldsymbol{M}) = (p+1) - 1 = p$; and $\boldsymbol{M}\boldsymbol{\mu} = \beta_0(\boldsymbol{1} - \boldsymbol{1}) = \boldsymbol{0}$ since $\boldsymbol{H}\boldsymbol{1} = \boldsymbol{1}$, $\boldsymbol{P}_0\boldsymbol{1} = \boldsymbol{1}$ (*this is where H_0 is used*: a constant regression function is annihilated by centring). Therefore $\hat{\boldsymbol{y}} - \bar{y}\boldsymbol{1} = \boldsymbol{M}\boldsymbol{\varepsilon} = \sigma \boldsymbol{M}\boldsymbol{z}$ and $\mathrm{RegSS}/\sigma^2 = \boldsymbol{z}^\top \boldsymbol{M}\boldsymbol{z} \sim \chi^2_p$.

(d) *Independence of RegSS and RSS.* $\boldsymbol{M}(\boldsymbol{I}_n - \boldsymbol{H}) = (\boldsymbol{H} - \boldsymbol{P}_0)(\boldsymbol{I}_n - \boldsymbol{H}) = \boldsymbol{H} - \boldsymbol{H}^2 - \boldsymbol{P}_0 + \boldsymbol{P}_0\boldsymbol{H} = \boldsymbol{0}$, so Lemma 6.8 applies. □

Corollary 6.10 (The ANOVA table)**.** *Under the normal linear model with an intercept:*

Source	*Sum of squares*	*df*	*Mean square*	F
Regression	$\mathrm{RegSS} = \sum_i (\hat{y}_i - \bar{y})^2$	p	$\mathrm{MSR} = \mathrm{RegSS}/p$	$\mathrm{MSR}/\mathrm{MSE}$
Error	$\mathrm{RSS} = \sum_i (y_i - \hat{y}_i)^2$	$n - p - 1$	$\mathrm{MSE} = \mathrm{RSS}/(n - p - 1) = s^2$	
Total	$\mathrm{TSS} = \sum_i (y_i - \bar{y})^2$	$n - 1$		

$\mathrm{MSE} = s^2$ *is unbiased for σ^2;* $\mathrm{RSS}/\sigma^2 \sim \chi^2_{n-p-1}$*; under H_0,* $\mathrm{RegSS}/\sigma^2 \sim \chi^2_p$ *independently of RSS. Degrees of freedom add exactly as sums of squares do:* $p + (n - p - 1) = n - 1$.

Remark 6.11 (Why the denominator df is $n - p - 1$ and not $n - 2$). The only change from SLR is that $p + 1$ parameters are estimated instead of 2; every $n - 2$ becomes $n - p - 1$. The t- and F-tables, standard-error and confidence-interval formulas and the R^2 definitions are otherwise identical. Also: the total line above is *not* what R prints; `anova()` prints the sequential (Type I) breakdown, one line per predictor, then the residual line (Section 10).

6.4 Step 4: the global F-test

Theorem 6.12 (Global F-test)**.** *For $H_0 : \beta_1 = \cdots = \beta_p = 0$ versus H_1: at least one $\beta_j \neq 0$,*

$$F \;=\; \frac{\mathrm{MSR}}{\mathrm{MSE}} \;=\; \frac{\mathrm{RegSS}/p}{\mathrm{RSS}/(n-p-1)} \;\sim\; F_{p,\,n-p-1} \text{ under } H_0,$$

independently of σ^2. Reject at level α when $F > F_{p,n-p-1;1-\alpha}$, equivalently when $\Pr(F_{p,n-p-1} > F_{obs}) < \alpha$.

Proof. $U = \mathrm{RegSS}/\sigma^2 \sim \chi^2_p$, $V = \mathrm{RSS}/\sigma^2 \sim \chi^2_{n-p-1}$, $U \perp\!\!\!\perp V$ under H_0 (Theorem 6.9). By definition of the F distribution,

$$\frac{U/p}{V/(n-p-1)} = \frac{\mathrm{RegSS}/\sigma^2}{p} \cdot \frac{n-p-1}{\mathrm{RSS}/\sigma^2} = \frac{\mathrm{RegSS}/p}{\mathrm{RSS}/(n-p-1)} = \frac{\mathrm{MSR}}{\mathrm{MSE}} = F.$$

σ^2 cancels, so F is a pivot. □

Remark 6.13 (What the global test does and does not tell you). Rejecting H_0 says only that *some* linear combination of predictors helps explain y; with correlated predictors the global F can be highly significant while every individual t is insignificant. A large p-value does not prove the predictors are useless, only that this sample provides no evidence that they are useful.

6.5 The coefficient of determination

Definition 6.14.

$$R^2 = \frac{\mathrm{RegSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}}, \qquad R^2_{\mathrm{adj}} = 1 - (1-R^2)\frac{n-1}{n-p-1} = 1 - \frac{\mathrm{RSS}/(n-p-1)}{\mathrm{TSS}/(n-1)}.$$

R^2 is the proportion of variation of y about its mean accounted for by the regression. In MLR R^2 can never decrease when a predictor is added (RSS cannot increase), which is why the adjusted version, penalising the $p+1$ estimated parameters, is also reported. Neither should be the sole criterion for model choice (Section 13).

7 Inference on a single coefficient, and the partial F-test

7.1 The t-test and confidence interval for one slope

Theorem 7.1 (t-statistic)**.** *For $H_0 : \beta_j = d$ against $H_1 : \beta_j \neq d$,*

$$t(\hat\beta_j) = \frac{\hat\beta_j - d}{\widehat{\mathrm{se}}(\hat\beta_j)} \sim t_{n-p-1} \text{ under } H_0,$$

with $\widehat{\mathrm{se}}$ as in Remark 5.9. A $100(1-\alpha)\%$ confidence interval for β_j is $\hat\beta_j \pm t_{n-p-1;1-\alpha/2}\,\widehat{\mathrm{se}}(\hat\beta_j)$.

Proof. By Theorem 5.8, $\hat\beta_j \sim N(\beta_j, \sigma^2 v_j)$ with $v_j = [(\boldsymbol{X}^\top\boldsymbol{X})^{-1}]_{jj}$, so under H_0, $Z = (\hat\beta_j - d)/(\sigma\sqrt{v_j}) \sim N(0,1)$. By Theorem 6.9(a), $V = \mathrm{RSS}/\sigma^2 \sim \chi^2_{n-p-1}$ and RSS is independent of $\hat{\boldsymbol{\beta}}$, hence of Z. Therefore

$$t = \frac{Z}{\sqrt{V/(n-p-1)}} = \frac{(\hat\beta_j - d)/(\sigma\sqrt{v_j})}{\sqrt{\mathrm{RSS}/[\sigma^2(n-p-1)]}}$$

$$= \frac{\hat\beta_j - d}{\sqrt{\text{MSE}\ v_j}} = \frac{\hat\beta_j - d}{\widehat{\text{se}}(\hat\beta_j)} \sim t_{n-p-1},$$

by the definition of Student's t. The confidence interval is the usual inversion of this test. □

7.2 The general linear (partial) F-test

Testing several slopes jointly, e.g. $H_0 : \beta_2 = \beta_3 = 0$, with separate t-tests is wrong: each controls its own type I error, so the chance of at least one false rejection is inflated above α. The correct tool is a single partial F-test.

Theorem 7.2 (Partial F-test for nested models)**.** *Reduced model (R):* $\mathbb{E}[y] = \beta_0 + \beta_1 x_1 + \cdots + \beta_q x_q$ *($q < p$); full model (F):* $\mathbb{E}[y] = \beta_0 + \cdots + \beta_p x_p$*. For* $H_0 : \beta_{q+1} = \cdots = \beta_p = 0$*,*

$$F \;=\; \frac{[\text{RSS(R)} - \text{RSS(F)}]/(p-q)}{\text{RSS(F)}/(n-p-1)} \;\sim\; F_{p-q,\,n-p-1} \text{ under } H_0.$$

Proof. Let $\boldsymbol{H}_R$, $\boldsymbol{H}_F$ be the hat matrices; the columns of $\boldsymbol{X}_R$ are the first $q+1$ columns of $\boldsymbol{X}_F$, so $\text{col}(\boldsymbol{X}_R) \subseteq \text{col}(\boldsymbol{X}_F)$ and $\boldsymbol{H}_R\boldsymbol{H}_F = \boldsymbol{H}_F\boldsymbol{H}_R = \boldsymbol{H}_R$.

(a) *Algebra.* $\text{RSS(F)} = \|\boldsymbol{y} - \boldsymbol{H}_F\boldsymbol{y}\|^2$, $\text{RSS(R)} = \|\boldsymbol{y} - \boldsymbol{H}_R\boldsymbol{y}\|^2$; the difference of nested projections $\boldsymbol{D} := \boldsymbol{H}_F - \boldsymbol{H}_R$ is symmetric idempotent with $\text{rank}(\boldsymbol{D}) = (p+1) - (q+1) = p - q$, and

$$\text{RSS(R)} - \text{RSS(F)} = \boldsymbol{y}^\top(\boldsymbol{I}_n - \boldsymbol{H}_R)\boldsymbol{y} - \boldsymbol{y}^\top(\boldsymbol{I}_n - \boldsymbol{H}_F)\boldsymbol{y} = \boldsymbol{y}^\top\boldsymbol{D}\boldsymbol{y}.$$

Under H_0, $\boldsymbol{\mu} = \boldsymbol{X}_F\boldsymbol{\beta} = \boldsymbol{X}_R\boldsymbol{\beta}_R$ lies in $\text{col}(\boldsymbol{X}_R)$, so $\boldsymbol{D}\boldsymbol{\mu} = \boldsymbol{0}$ and $\text{RSS(R)} - \text{RSS(F)} = \boldsymbol{\varepsilon}^\top\boldsymbol{D}\boldsymbol{\varepsilon} = \sigma^2\boldsymbol{z}^\top\boldsymbol{D}\boldsymbol{z}$, giving $[\text{RSS(R)} - \text{RSS(F)}]/\sigma^2 \sim \chi^2_{p-q}$.

(b) *Independence.* $\boldsymbol{D}(\boldsymbol{I}_n - \boldsymbol{H}_F) = \boldsymbol{H}_F - \boldsymbol{H}_F^2 - \boldsymbol{H}_R + \boldsymbol{H}_R\boldsymbol{H}_F = \boldsymbol{0}$, so the numerator is independent of RSS(F).

(c) Divide the independent chi-squares by their df and cancel σ^2. □

Corollary 7.3 (Partial F for one predictor is t^2)**.** *For* $H_0 : \beta_j = 0$ *against the full model (*$q = p-1$*, omitted predictor* x_j*),* $F_{1,n-p-1} = t(\hat\beta_j)^2$ *and both give the same* p*-value.*

Proof sketch. Dropping x_j increases RSS by $\text{RSS(R)} - \text{RSS(F)} = \hat\beta_j^2/\widehat{\text{se}}(\hat\beta_j)^2 \cdot \text{MSE}$. Substituting with $p - q = 1$ gives $F = \hat\beta_j^2/\widehat{\text{se}}(\hat\beta_j)^2 = t(\hat\beta_j)^2$. □

Remark 7.4 (Special case: the global test). With $q = 0$ the reduced model is intercept-only, $\text{RSS(R)} = \text{TSS}$, and Theorem 7.2 becomes Theorem 6.12: the global F-test is the partial F-test of "all slopes are zero".

8 Diagnostics: leverage, residuals and influence

8.1 Leverage

The i-th fitted value is $\hat y_i = \boldsymbol{h}_i^\top\boldsymbol{y} = \sum_{j=1}^n h_{ij}y_j$: h_{ij} measures how much observation j pulls on the i-th fitted value.

Definition 8.1 (Leverage)**.** $h_{ii} = \boldsymbol{x}_i^\top(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{x}_i$, the i-th diagonal element of $\boldsymbol{H}$, where $\boldsymbol{x}_i^\top = (1, x_{i1}, \ldots, x_{ip})$ is the i-th row of $\boldsymbol{X}$.

Proposition 8.2 (Properties of leverage)**.** $0 \le h_{ii} \le 1$, $\sum_{i=1}^{n} h_{ii} = p+1 = k$, $\bar{h} = k/n$. *Rule of thumb: observation i is a high-leverage point if*

$$h_{ii} > c\,\bar{h} = c\,\frac{p+1}{n}, \qquad c \in [2,4].$$

Proof. $\boldsymbol{H}$ idempotent and symmetric gives $h_{ii} = [\boldsymbol{H}^2]_{ii} = \sum_j h_{ij}h_{ji} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j\neq i} h_{ij}^2 \ge h_{ii}^2$, so $0 \le h_{ii} \le 1$; and $\sum_i h_{ii} = \operatorname{tr}(\boldsymbol{H}) = k$. □

Leverage depends only on the $\boldsymbol{x}$'s, not on $\boldsymbol{y}$: a high-leverage point has unusual predictor values (far from the centre of the x-cloud). In the fuel example the largest leverage is $h_{66} = 0.5838$ against the cut-off $2k/n = 0.75$, so no observation is flagged.

8.2 Residual variance and standardised residuals

Proposition 8.3 (Residuals are correlated, and shrunk)**.** *Under the normal linear model $\boldsymbol{e} \sim N_n(\boldsymbol{0}, \sigma^2(\boldsymbol{I}_n - \boldsymbol{H}))$, so*

$$\mathbb{E}[e_i] = 0, \qquad \operatorname{Var}(e_i) = \sigma^2(1-h_{ii}) \le \sigma^2, \qquad \operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij} \ (i \neq j).$$

Every residual has variance smaller than σ^2, and residuals are mutually correlated even when the errors are independent.

Proof. $\boldsymbol{e} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{\varepsilon}$ is a linear transformation of a normal vector, with mean $\boldsymbol{0}$ and covariance $(\boldsymbol{I}_n - \boldsymbol{H})\sigma^2\boldsymbol{I}_n(\boldsymbol{I}_n - \boldsymbol{H})^\top = \sigma^2(\boldsymbol{I}_n - \boldsymbol{H})$ by symmetry and idempotence. □

Because $\operatorname{Var}(e_i)$ depends on i, raw residuals are not comparable across observations.

Definition 8.4 (Standardised and studentised residuals)**.**

$$r_i = \frac{e_i}{s\sqrt{1-h_{ii}}}, \qquad r_i^* = \frac{e_i}{s_{(i)}\sqrt{1-h_{ii}}}$$

where $s = \sqrt{\mathrm{MSE}}$ uses all n observations and $s_{(i)}$ is the residual standard error of the model refitted with observation i deleted. r_i is the standardised, r_i^* the (externally) studentised residual.

The studentised version is preferred for outliers: when y_i is unusually large, s is inflated by that very observation and the inflation appears in both numerator and denominator of r_i. Computing $s_{(i)}$ does not require n regressions:

$$s_{(i)}^2 = \frac{\mathrm{RSS} - \dfrac{e_i^2}{1-h_{ii}}}{n-p-2}, \qquad r_i^* \sim t_{n-p-2} \text{ under the model.}$$

Rule of thumb: examine any observation with $|r_i| > 2$ (or $|r_i^*| > 2$) more closely; it is a flag for inspection, not automatic deletion.

8.3 Delete-one identities

Proposition 8.5 (Delete-one identities)**.** *With $\boldsymbol{h}_i^\top$ the i-th row of $\boldsymbol{H}$,*

$$\hat{\boldsymbol{y}}_{(i)} = \hat{\boldsymbol{y}} - \frac{\boldsymbol{h}_i e_i}{1-h_{ii}}, \qquad \hat{\boldsymbol{\beta}}_{(i)} = \hat{\boldsymbol{\beta}} - \frac{(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{x}_i e_i}{1-h_{ii}}, \qquad s_{(i)}^2 = \frac{\mathrm{RSS} - e_i^2/(1-h_{ii})}{n-k-1}.$$

Consequently $r_i^* \sim t_{n-k-1}$ under the model.

Proof sketch. Let $\boldsymbol{X}_{(i)}$ be $\boldsymbol{X}$ with row i removed. The block-inverse identity gives

$$(\boldsymbol{X}_{(i)}^\top \boldsymbol{X}_{(i)})^{-1} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} + \frac{(\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_i \boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1}}{1 - h_{ii}};$$

substituting into $\hat{\boldsymbol{\beta}}_{(i)} = (\boldsymbol{X}_{(i)}^\top \boldsymbol{X}_{(i)})^{-1} \boldsymbol{X}_{(i)}^\top \boldsymbol{y}_{(i)}$ and simplifying gives the coefficient formula; premultiplying by $\boldsymbol{X}$ gives the fitted-value formula. The residual sum of squares of the deleted fit is $\mathrm{RSS}_{(i)} = \mathrm{RSS} - e_i^2/(1 - h_{ii})$. □

8.4 Influence and Cook's distance

An observation is *influential* if removing it changes the fitted coefficients substantially. Let $\hat{y}_{j(i)}$ be the fitted values without observation i.

Definition 8.6 (Cook's distance)**.**

$$\boxed{D_i = \frac{\sum_{j=1}^{n} (\hat{y}_j - \hat{y}_{j(i)})^2}{k\, s^2} = \frac{e_i^2}{k\, s^2} \cdot \frac{h_{ii}}{(1 - h_{ii})^2}}$$

The second form needs only e_i, h_{ii} and s^2. A common cut-off is $D_i > 4/(n-k) = 4/(n-p-1)$.

Derivation of the closed form. From Proposition 8.5, $\hat{y}_j - \hat{y}_{j(i)} = h_{ji} e_i/(1-h_{ii})$, so

$$\sum_j (\hat{y}_j - \hat{y}_{j(i)})^2 = \frac{e_i^2}{(1-h_{ii})^2} \sum_j h_{ji}^2 = \frac{e_i^2}{(1-h_{ii})^2} h_{ii},$$

using $h_{ii} = \sum_j h_{ij}^2$ (Proposition 8.2). Divide by ks^2. □

Outlier, leverage and influence are three different things:

- an outlier with small leverage: a large residual at an ordinary $\boldsymbol{x}$; moves the fit little but inflates MSE;
- a high-leverage point that is not an outlier: an unusual $\boldsymbol{x}$ close to the fitted surface; does not change the fit much;
- an influential point: usually an outlier *with* high leverage, which moves the surface and inflates MSE.

Unusual observations should not be deleted automatically. Ask *why* they are unusual (recording error, missing predictor, a genuinely different subpopulation); then correct the data, refit without them as a sensitivity check, extend the model, or use a robust method.

8.5 Collinearity and the variance inflation factor

If two predictors are highly correlated the data cannot separate their effects: the likelihood is nearly flat along the ridge $\beta_1 + \beta_2 = \text{const}$ and individual slopes are imprecise even though predictions may be fine.

Definition 8.7 (Variance inflation factor)**.** Regress x_j on all other predictors, with coefficient of determination R_j^2. Then $\mathrm{VIF}_j = \dfrac{1}{1 - R_j^2}$.

Proposition 8.8 (Interpretation)**.** $\mathrm{Var}(\hat{\beta}_j) = \sigma^2\, \mathrm{VIF}_j\, /[(n-1)\widehat{\mathrm{Var}}(x_j)]$*, so* VIF_j *is exactly the factor by which* $\mathrm{Var}(\hat{\beta}_j)$ *is inflated relative to the case where* x_j *is orthogonal to the other predictors.* $\mathrm{VIF}_j = 1$ *means no collinearity; values above about 5 (some texts say 10) warn that individual slopes are poorly determined.* VIF_j *is the* j*-th diagonal element of* $(\boldsymbol{X}^\top \boldsymbol{X})^{-1}$ *rescaled, and involves only the predictors, never the response.*

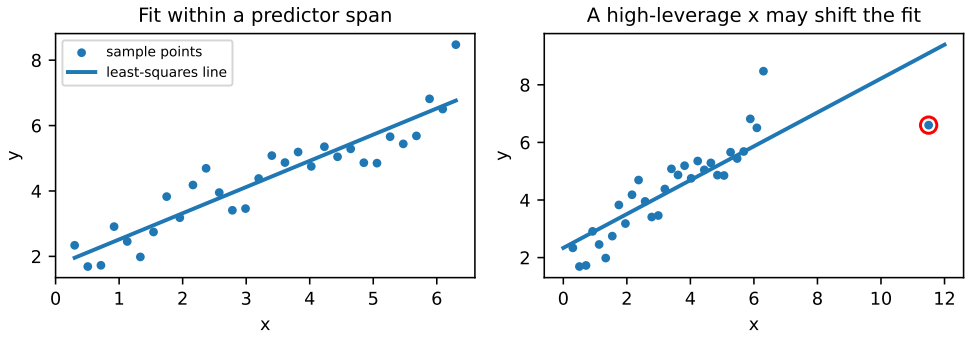


Figure 1: Projection geometry and leverage in two regressions (illustrative). R: `lm()`, `hatvalues()`, `plot()` or `ggplot2::geom_point()` plus `geom_smooth()`.

9 Worked example computed by hand: the fuel data

Hand first, R second

Every number here was recomputed independently; the formulas and the software are the same object.

Every number in the ANOVA table from first principles; the example is small enough for a calculator and a 3×3 inverse, but the decomposition, degrees of freedom, F-test and partial F-test are identical in larger problems.

Example 9.1 (Fuel use, Bowerman & Schafer 1990)**.** Weekly fuel consumption y is explained by average hourly temperature x_1 (řF) and the chill index x_2:

Obs i	Temperature x_{i1}	Chill index x_{i2}	Fuel use y_i	Fitted $\hat{y}_i$
1	28.0	18	12.4	12.0732583
2	28.0	14	11.7	11.7432784
3	32.5	24	12.4	12.1631658
4	39.0	22	10.8	11.4130857
5	45.9	8	9.4	9.6370603
6	57.8	16	9.5	9.2258550
7	58.1	1	8.0	7.9614262
8	62.5	0	7.5	7.4828702

$n = 8$, $p = 2$, $k = 3$, error df $n - k = 5$. Model: $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$, $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

Raw sums (everything else is built from these).

$n = 8$	$\sum_i y_i = 81.70$	$\sum_i x_{i1} = 351.80$	$\sum_i x_{i2} = 103.00$
$\sum_i x_{i1}^2 = 16874.76$	$\sum_i x_{i2}^2 = 1901.00$	$\sum_i x_{i1}x_{i2} = 3884.10$	
$\sum_i x_{i1}y_i = 3413.11$	$\sum_i x_{i2}y_i = 1157.40$	$\sum_i y_i^2 = 859.91$	$\bar{y} = 10.2125$

Step 1: $\boldsymbol{X}^\top\boldsymbol{X}$ and $\boldsymbol{X}^\top\boldsymbol{y}$

$$\boldsymbol{X}^\top\boldsymbol{X} = \begin{pmatrix} 8 & 351.80 & 103.00 \\ 351.80 & 16874.76 & 3884.10 \\ 103.00 & 3884.10 & 1901.00 \end{pmatrix}, \qquad \boldsymbol{X}^\top\boldsymbol{y} = \begin{pmatrix} 81.70 \\ 3413.11 \\ 1157.40 \end{pmatrix}.$$

Every entry is a raw sum: the $(1,1)$ entry is n, the $(1,2)$ entry is $\sum_i x_{i1}$, and so on.

Step 2: the inverse of $\boldsymbol{X}^\top\boldsymbol{X}$

Adjugate formula $(\boldsymbol{X}^\top\boldsymbol{X})^{-1} = \operatorname{adj}(\boldsymbol{X}^\top\boldsymbol{X})/\det(\boldsymbol{X}^\top\boldsymbol{X})$. **Determinant**, expanding along the first row with cofactors C_{1j}:

$$\begin{aligned} C_{11} &= 1901 \times 16874.76 - 3884.10^2 = 16\,992\,685.95, \\ C_{12} &= -(351.80 \times 1901 - 3884.10 \times 103) = -268\,709.50, \\ C_{13} &= 351.80 \times 3884.10 - 16874.76 \times 103 = -371\,673.90, \end{aligned}$$

$$\begin{aligned} \det(\boldsymbol{X}^\top\boldsymbol{X}) &= 8C_{11} + 351.80\,C_{12} + 103\,C_{13} \\ &= 135\,941\,487.6 - 94\,532\,002.1 - 38\,282\,411.7 = 3\,127\,073.8. \end{aligned}$$

A slicker route: because the first column is constant, the determinant equals n times the determinant of the *centred* cross-product matrix. Writing $S_{jl} = \sum_i x_{ij}x_{il} - n\bar{x}_j\bar{x}_l$,

$$\boldsymbol{S} = \begin{pmatrix} S_{11} & S_{12} \\ S_{12} & S_{22} \end{pmatrix} = \begin{pmatrix} 1404.355 & -645.325 \\ -645.325 & 574.875 \end{pmatrix}, \qquad \det(\boldsymbol{S}) = 390\,884.225,$$

and $8 \times 390\,884.225 = 3\,127\,073.8$. The centred quantities are also what the slope formulas and the partial F-test need.

Remaining cofactors and the inverse.

$$C_{22} = 8 \times 1901 - 103^2 = 4599.00, \qquad C_{23} = -(8 \times 3884.10 - 103 \times 351.80) = 5162.60,$$

$$C_{33} = 8 \times 16874.76 - 351.80^2 = 11234.84.$$

The cofactor matrix is symmetric, so the adjugate equals it:

$$(\boldsymbol{X}^\top\boldsymbol{X})^{-1} = \frac{1}{3\,127\,073.8} \begin{pmatrix} 16\,992\,685.95 & -268\,709.50 & -371\,673.90 \\ -268\,709.50 & 4599.00 & 5162.60 \\ -371\,673.90 & 5162.60 & 11234.84 \end{pmatrix}$$

Evaluating the nine quotients gives

$$(\boldsymbol{X}^\top\boldsymbol{X})^{-1} = \begin{pmatrix} 5.434053 & -0.085930 & -0.118857 \\ -0.085930 & 0.001471 & 0.001651 \\ -0.118857 & 0.001651 & 0.003593 \end{pmatrix}.$$

Step 3: the least squares estimate

Multiplying the inverse by $\boldsymbol{X}^\top\boldsymbol{y}$ row by row,

$$\begin{aligned} \hat{\beta}_0 &= 5.434053(81.70) - 0.085930(3413.11) - 0.118857(1157.40) = 13.108737, \\ \hat{\beta}_1 &= -0.085930(81.70) + 0.001471(3413.11) + 0.001651(1157.40) = -0.090014, \\ \hat{\beta}_2 &= -0.118857(81.70) + 0.001651(3413.11) + 0.003593(1157.40) = 0.082495. \end{aligned}$$

An exact cross-check. With the integer adjugate instead of the rounded inverse,

$$\operatorname{adj}(\boldsymbol{X}^\top\boldsymbol{X})\,\boldsymbol{X}^\top\boldsymbol{y} = \begin{pmatrix} 40\,991\,988.71 \\ -281\,480.02 \\ 257\,967.872 \end{pmatrix},$$

and dividing by $3\,127\,073.8$ gives $\hat{\boldsymbol{\beta}} = (13.1087372,\ -0.0900139,\ 0.0824950)^\top$. The 6-decimal inverse instead gives 13.108738, -0.0900144, 0.0824940: the last digits move because of rounding, a lesson in how many significant figures to carry.

$$\widehat{\text{Fuel}} \;=\; 13.1087 \;-\; 0.09001\,\text{Temp} \;+\; 0.08249\,\text{Chill}$$

Interpretation. Holding the chill index fixed, a one-degree increase in temperature is associated with a decrease of 0.09 units of fuel use; holding temperature fixed, a one-unit increase in chill index with an increase of 0.082 units.

A second route (exam route, 2×2 inverse only). With the centred sums and

$$S_{1y} = \sum_i x_{i1} y_i - n\bar{x}_1 \bar{y} = -179.6475, \qquad S_{2y} = \sum_i x_{i2} y_i - n\bar{x}_2 \bar{y} = 105.5125,$$

$$S_{yy} = \sum_i y_i^2 - n\bar{y}^2 = 25.54875,$$

$$\begin{aligned} \hat{\beta}_1 &= \frac{S_{22} S_{1y} - S_{12} S_{2y}}{S_{11} S_{22} - S_{12}^2} = \frac{574.875(-179.6475) + 645.325(105.5125)}{390\,884.225} \\ &= -0.090014, \\ \hat{\beta}_2 &= \frac{S_{11} S_{2y} - S_{12} S_{1y}}{S_{11} S_{22} - S_{12}^2} = \frac{1404.355(105.5125) + 645.325(-179.6475)}{390\,884.225} \\ &= 0.082495, \end{aligned}$$

and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}_1 - \hat{\beta}_2 \bar{x}_2 = 10.2125 + 3.95836 - 1.06212 = 13.108737$.

Step 4: fitted values, residuals and the orthogonality checks

i	1	2	3	4	5	6	7	8
e_i	0.326742	−0.043278	0.236834	−0.613086	−0.237060	0.274145	0.038574	0.017130
e_i^2	0.106760	0.001873	0.056090	0.375874	0.056198	0.075155	0.001488	0.000293

Checkpoint 9.2 (The checks that must come out exactly right)**.**

$$\begin{aligned} \textstyle\sum_i e_i &= -6 \times 10^{-13} \approx 0 && \text{(first normal equation)} \\ \textstyle\sum_i x_{i1} e_i &= -2.6 \times 10^{-11} \approx 0 && \text{(second normal equation)} \\ \textstyle\sum_i x_{i2} e_i &= -9 \times 10^{-12} \approx 0 && \text{(third normal equation)} \\ \textstyle\sum_i \hat{y}_i e_i &= -6.4 \times 10^{-12} \approx 0 && \text{(residuals orthogonal to the fit)} \\ \textstyle\sum_i \hat{y}_i &= \textstyle\sum_i y_i = 81.70 && (\text{so } \bar{\hat{y}} = \bar{y}) \end{aligned}$$

and the two identities that make the ANOVA decomposition work: $\sum_i \hat{y}_i^2 = 859.236268$, $\sum_i y_i^2 - \sum_i e_i^2 = 859.91 - 0.673732 = 859.236268$. Tiny non-zero values are floating-point round-off.

Step 5: the three sums of squares

$$\begin{aligned} \text{TSS} &= \textstyle\sum_i y_i^2 - n\bar{y}^2 = 859.91 - 8(10.2125)^2 = 859.91 - 834.36125 = 25.54875, \\ \text{RegSS} &= \textstyle\sum_i \hat{y}_i^2 - n\bar{y}^2 = 859.236268 - 834.36125 = 24.875018, \\ \text{RSS} &= \textstyle\sum_i e_i^2 = 0.673732. \end{aligned}$$

Checks: $\text{RegSS} + \text{RSS} = 24.875018 + 0.673732 = 25.548750 = \text{TSS}$; the "$\hat{\boldsymbol{\beta}}$ route" $\hat{\boldsymbol{\beta}}^\top \boldsymbol{X}^\top \boldsymbol{y} - n\bar{y}^2 = 859.236268 - 834.36125 = 24.875018$ agrees; $R^2 = 24.875018/25.54875 = 0.973630$: the two predictors explain 97.4% of the variation in weekly fuel use.

Step 6: the ANOVA table and the global F-test

With $p = 2$, $n - p - 1 = 5$, $n - 1 = 7$:

Source	SS	df	MS	F	p-value
Regression	24.875018	2	12.437509	92.3031	0.00011293
Error	0.673732	5	0.134746		
Total	25.548750	7			

$$\mathrm{MSR} = \frac{24.875018}{2} = 12.437509, \qquad \mathrm{MSE} = \frac{0.673732}{5} = 0.1347464,$$

$$s = \sqrt{\mathrm{MSE}} = 0.367078, \qquad F = \frac{\mathrm{MSR}}{\mathrm{MSE}} = \frac{12.437509}{0.1347464} = 92.3031,$$

$\Pr(F_{2,5} > 92.3031) = 1.129 \times 10^{-4}$. Critical value $F_{2,5;0.95} = 5.7861$; since $92.30 \gg 5.79$ we reject $H_0 : \beta_1 = \beta_2 = 0$ decisively. Adjusted $R^2_{\text{adj}} = 1 - (1 - 0.973630)\frac{7}{5} = 0.963081$.

Step 7: standard errors, t-tests and confidence intervals

$$\widehat{\mathrm{se}}(\hat\beta_0) = \sqrt{0.1347464 \times 5.434053} = 0.855698,$$
$$\widehat{\mathrm{se}}(\hat\beta_1) = \sqrt{0.1347464 \times 0.001470704} = 0.0140774,$$
$$\widehat{\mathrm{se}}(\hat\beta_2) = \sqrt{0.1347464 \times 0.003592765} = 0.0220025.$$

With $t_{5;0.975} = 2.570582$:

Parameter	Estimate	Std. error	t	95% CI
β_0	13.108737	0.855698	15.31935	(10.9091, 15.3084)
β_1 (Temp)	-0.090014	0.014077	-6.3942	(-0.12620, -0.05383)
β_2 (Chill)	0.082495	0.022003	3.7493	(0.02594, 0.13905)

p-values 2.15×10^{-5}, 0.001386, 0.013303. Both slopes are significant at 5%; the interval for β_1 lies entirely below zero, so temperature has a significantly negative effect once the chill index is controlled for.

Step 8: the partial F-test, and $t^2 = F$

Is the chill index needed *given* temperature? Test $H_0 : \beta_2 = 0$ against the full model. The reduced model $y \sim$ Temp (from the same raw sums):

$$\widehat{\mathrm{Fuel}} = 15.8379 - 0.12792\,\mathrm{Temp}, \qquad \mathrm{RSS(R)} = 2.567934, \qquad R^2 = 0.899489.$$

$$F = \frac{[\mathrm{RSS(R)} - \mathrm{RSS(F)}]/(p-q)}{\mathrm{RSS(F)}/(n-p-1)} = \frac{(2.567934 - 0.673732)/1}{0.673732/5}$$
$$= \frac{1.894202}{0.1347464} = 14.0575,$$

$\Pr(F_{1,5} > 14.0575) = 0.013303$: reject $H_0 : \beta_2 = 0$ at 5%; the chill index adds significantly to the model already containing temperature.

Checkpoint 9.3 ($t^2 = F$)**.** From Step 7, $t(\hat\beta_2) = 3.749337$ and $t(\hat\beta_2)^2 = 14.0575 = F$. The partial F and the t-test of the same slope are the same test; the quickest check that both computations are right.

Remark 9.4 (A no-refit formula for the partial F). The extra sum of squares from adding x_2 to a model containing x_1 can be written from full-model quantities:

$$\mathrm{RSS(R)} - \mathrm{RSS(F)} = \hat\beta_2^2\Big(S_{22} - \frac{S_{12}^2}{S_{11}}\Big) = 0.082495^2 \times 278.33719 = 1.894202,$$

the MLR analogue of the SLR identity $\mathrm{RSS(R)} - \mathrm{RSS(F)} = \hat\beta_1^2 S_{xx}$.

Step 9: residual diagnostics for this data set

Leverages (diagonal of $\boldsymbol{H}$):

$$h_{ii} = (0.3244,\ 0.4455,\ 0.3418,\ 0.3107,\ 0.1848,\ 0.5838,\ 0.3712,\ 0.4377), \quad \textstyle\sum_i h_{ii} = 3 = k.$$

$\bar{h} = 3/8 = 0.375$; with $c = 2$ the cut-off is 0.75: no high-leverage point, though observation 6 ($h_{66} = 0.5838$, temperature 57.8 with chill 16) is the most extreme.

i	e_i	h_{ii}	r_i (standardised)	r_i^* (studentised)	D_i (Cook)
1	0.326742	0.3244	1.0829	1.1071	0.1877
2	−0.043278	0.4455	−0.1583	−0.1420	0.0067
3	0.236834	0.3418	0.7953	0.7611	0.1095
4	−0.613086	0.3107	−2.0116	**−4.1204**	**0.6079**
5	−0.237060	0.1848	−0.7153	−0.6753	0.0387
6	0.274145	0.5838	1.1577	1.2103	0.6267
7	0.038574	0.3712	0.1325	0.1187	0.0035
8	0.017130	0.4377	0.0622	0.0557	0.0010

Worked detail for observation 4 (temperature 39řF, chill 22, fuel 10.8):

$$\begin{aligned}
e_4 &= -0.613086, \qquad h_{44} = 0.310660,\\
r_4 &= \frac{-0.613086}{0.367078\sqrt{1-0.310660}} = -2.0116,\\
s_{(4)} &= \sqrt{\frac{0.673732 - \frac{(-0.613086)^2}{1-0.310660}}{5}} = \sqrt{\frac{0.128466}{5}} = 0.179210,\\
r_4^* &= \frac{-0.613086}{0.179210\sqrt{1-0.310660}} = -4.1204,\\
D_4 &= \frac{(-0.613086)^2}{3(0.1347464)} \cdot \frac{0.310660}{(1-0.310660)^2} = 0.6079, \qquad \text{cut-off } \frac{4}{n-k} = 0.8.
\end{aligned}$$

Observation 4 is an outlier by the studentised rule ($|r^*| = 4.12 > 2$) even though its standardised residual is only just over 2 in absolute value, exactly the case that motivates r^*. Its leverage is unremarkable, but its Cook distance is among the largest, so it is also highly influential; refitting without it changes the coefficients noticeably. It should not be deleted automatically: the right conclusion is to investigate why that week behaved as it did.

Step 10: prediction for a new observation

For a week with temperature 30řF and chill index 10, $\boldsymbol{x}_0^\top = (1, 30, 10)$:

$$\hat{y}_0 = \boldsymbol{x}_0^\top \hat{\boldsymbol{\beta}} = 13.108737 - 0.090014(30) + 0.082495(10) = 11.233271,$$

and $\boldsymbol{x}_0^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_0 = 0.574589$, giving

$$\widehat{\text{se}}(\text{mean response}) = \sqrt{\text{MSE}\ \boldsymbol{x}_0^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_0} = \sqrt{0.1347464 \times 0.574589} = 0.278251,$$

$$\widehat{\text{se}}(\text{individual}) = \sqrt{\text{MSE}\,(1 + \boldsymbol{x}_0^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_0)} = 0.460619.$$

95% CI for the mean response: $(10.5180,\ 11.9485)$;
95% PI for a new week: $(10.0492,\ 12.4173)$.

The prediction interval is wider, as it must be: it includes the variability σ^2 of a single new observation on top of the uncertainty in the estimated mean.

9.1 What changes without an intercept

Without an intercept none of the ANOVA table applies as written: $\mathbf{1}$ is no longer a column of $\boldsymbol{X}$, the fitted values need not sum to the observed ones, and the proof of Theorem 6.2 fails at $\sum_i \hat{y}_i = \sum_i y_i$. The "total" becomes the uncorrected $\sum_i y_i^2$ rather than $\sum_i (y_i - \bar{y})^2$, and the regression sum of squares is no longer a sum of squared deviations from a mean. For the fuel data the no-intercept fit has $\mathrm{RSS} = 32.296350$, exceeding $\sum_i (y_i - \bar{y})^2 = 25.54875$; the uncorrected total is 859.91 and the printed R^2 is not comparable with the with-intercept value. Retain the intercept unless there is a strong substantive reason, and then re-derive.

10 The same example in R: side-by-side verification

```
Fuel  <- c(12.4, 11.7, 12.4, 10.8, 9.4, 9.5, 8.0, 7.5)
Temp  <- c(28.0, 28.0, 32.5, 39.0, 45.9, 57.8, 58.1, 62.5)
Chill <- c(18, 14, 24, 22, 8, 16, 1, 0)
fit <- lm(Fuel ~ Temp + Chill)
summary(fit)
anova(fit)
```

```
Call:  lm(formula = Fuel ~ Temp + Chill)
Residuals:
       1        2        3        4        5        6        7        8
0.326742 -0.043278 0.236834 -0.613086 -0.237060 0.274145 0.038574 0.017130
Coefficients:
             Estimate Std. Error t value  Pr(>|t|)
(Intercept) 13.108737   0.855698 15.3193 0.0000215 ***
Temp        -0.090014   0.014077 -6.3942  0.001386 **
Chill        0.082495   0.022003  3.7493  0.013303 *
Residual standard error: 0.36708 on 5 degrees of freedom
Multiple R-squared: 0.97363,   Adjusted R-squared: 0.96308
F-statistic: 92.303 on 2 and 5 DF,  p-value: 0.00011293

> anova(fit)
Response: Fuel
          Df   Sum Sq  Mean Sq  F value      Pr(>F)
Temp       1 22.98082 22.98082 170.5486 0.000046966 ***
Chill      1  1.89420  1.89420  14.0575    0.013303 *
Residuals  5  0.67373  0.13475
```

Remark 10.1 (How to read R's `anova()` output). It is a *sequential* (Type I) table: line 1 is the reduction in RSS from fitting Temp alone, line 2 the *additional* reduction from adding Chill after Temp. Hence

$$\underbrace{22.98082}_{\text{Temp}} + \underbrace{1.89420}_{\text{Chill}} = 24.87502 = \mathrm{RegSS}\,.$$

The residual line is $\mathrm{RSS} = 0.67373$ with 5 df. R prints no total line; construct it as $\mathrm{RegSS} + \mathrm{RSS} = 25.54875$ on 7 df. Two traps:

- The `F value` column does *not* test the whole model. Row 2 tests $\beta_2 = 0$ given Temp, i.e. the partial F: $1.89420/0.13475 = 14.0575$. The global F-test of $H_0 : \beta_1 = \beta_2 = 0$ is 92.303, on the last line of `summary()`.
- The sequential breakdown depends on the order of terms. `lm(Fuel ~ Chill + Temp)` gives different first and second lines but the same RegSS, the same $F = 92.303$ and the same fitted model. With $p = 2$ the two lines are the two partial F's; with correlated predictors the order genuinely matters, so fix an interpretable order or use Type II/III sums of squares (`car::Anova`).

10.1 The hand-versus-R comparison table

```
TSS                          hand = 25.548750    R = 25.548750
Reg SS                       hand = 24.875018    R = 24.875018
RSS                          hand = 0.673732     R = 0.673732
MSE = s^2                    hand = 0.134746     R = 0.134746
GLOBAL F (all slopes 0)      hand = 92.303086    R = 92.303086
R^2                          hand = 0.973630     R = 0.973630
adjusted R^2                 hand = 0.963081     R = 0.963081
beta0                        hand = 13.108737    R = 13.108737
beta1                        hand = -0.090014    R = -0.090014
beta2                        hand = 0.082495     R = 0.082495
se(beta0)                    hand = 0.855698     R = 0.855698
se(beta1)                    hand = 0.014077     R = 0.014077
se(beta2)                    hand = 0.022003     R = 0.022003
t(beta1)                     hand = -6.394229    R = -6.394229
t(beta2)                     hand = 3.749337     R = 3.749337
CI lower beta1               hand = -0.126201    R = -0.126201
CI upper beta1               hand = -0.053827    R = -0.053827
t(beta2)^2 vs partial F      hand = 14.057530    R = 14.057530
centred-formula partial F    hand = 14.057530    R = 14.057530
sum h_ii                     hand = 3.000000     R = 3.000000
max |r_std - rstandard|      hand = 0.000000     R = 0.000000
max |r_stud - rstudent|      hand = 0.000000     R = 0.000000
```

There is no DIFFERENT line: every hand-computed quantity reproduces R exactly to the printed precision. The formula on the board and the black box in R are the same object.

10.2 The pieces R computes for you

```
summary(fit)$fstatistic        # value 92.303086, numdf 2, dendf 5   (global F)
fit_R <- lm(Fuel ~ Temp)       # partial F: full vs reduced model
anova(fit_R, fit)
  Res.Df      RSS Df Sum of Sq       F   Pr(>F)
1      6 2.567934
2      5 0.673732  1    1.8942 14.0575 0.013303 *
hatvalues(fit); rstandard(fit); rstudent(fit); cooks.distance(fit)
predict(fit, newdata = data.frame(Temp = 30, Chill = 10), interval = "confidence")
predict(fit, newdata = data.frame(Temp = 30, Chill = 10), interval = "prediction")
       fit      lwr      upr                fit      lwr      upr
1 11.23327 10.51800 11.94854        1 11.23327 10.04921 12.41733
```

11 The hat matrix in full

11.1 What H is, and why it is called the hat matrix

One matrix, every diagnostic

Leverage, residual covariance, standard errors, studentised residuals and Cook's distance are all traces and quadratic forms in H.

With $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{y}$,

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{H}\boldsymbol{y}, \qquad \boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top,$$

so $\boldsymbol{H}$ maps observed responses to fitted values. It is $n \times n$ (square, in general not of full rank), built entirely from the design matrix: $\boldsymbol{y}$ does not appear in it.

Remark 11.1 (Geometric content). $\boldsymbol{H}$ is the orthogonal projection of $\mathbb{R}^n$ onto $\mathcal{C}(\boldsymbol{X})$ and $\boldsymbol{I}_n - \boldsymbol{H}$ the projection onto $\mathcal{C}(\boldsymbol{X})^\perp$: $\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y} \in \mathcal{C}(\boldsymbol{X})$, $\boldsymbol{e} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y} \in \mathcal{C}(\boldsymbol{X})^\perp$, $\hat{\boldsymbol{y}}^\top \boldsymbol{e} = 0$. This single fact is the source of $\mathrm{TSS} = \mathrm{RegSS} + \mathrm{RSS}$ and of $\boldsymbol{X}^\top \boldsymbol{e} = \boldsymbol{0}$.

Remark 11.2 (How to read the entries). Row i gives the weights with which observations enter the i-th fitted value, $\hat{y}_i = \sum_{j=1}^{n} h_{ij} y_j = h_{i1}y_1 + \cdots + h_{ii}y_i + \cdots + h_{in}y_n$. h_{ij} measures how strongly observation j pulls on $\hat{y}_i$, and h_{ii} (leverage) how strongly observation i pulls on its own fitted value. Entries can be negative: if j sits on the opposite side of the x-cloud from i, increasing y_j pulls $\hat{y}_i$ down.

11.2 The complete H for the fuel example

All 64 entries ($\hat{y} = 13.1087 - 0.09001\,\mathrm{Temp} + 0.08249\,\mathrm{Chill}$):

	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8
obs 1	**0.32436**	0.35620	0.20895	0.12715	0.13487	−0.10771	0.00719	−0.05100
obs 2	0.35620	**0.44553**	0.12484	0.02887	0.19222	−0.24392	0.08457	0.01169
obs 3	0.20895	0.12484	**0.34180**	0.30944	0.02537	0.21131	−0.10362	−0.11809
obs 4	0.12715	0.02887	0.30944	**0.31066**	0.02013	0.30890	−0.05733	−0.04781
obs 5	0.13487	0.19222	0.02537	0.02013	**0.18485**	0.00807	0.22155	0.21294
obs 6	−0.10771	−0.24392	0.21131	0.30890	0.00807	**0.58383**	0.08071	0.15882
obs 7	0.00719	0.08457	−0.10362	−0.05733	0.22155	0.08071	**0.37123**	0.39571
obs 8	−0.05100	0.01169	−0.11809	−0.04781	0.21294	0.15882	0.39571	**0.43774**

Three things to notice. **The matrix is symmetric**: entries $(1,2)$ and $(2,1)$ are both 0.35620. **The diagonal is the largest entry in its row**, which is what makes $h_{ii} \geq h_{ij}^2$ and $0 \leq h_{ii} \leq 1$. **Several entries are negative**, the largest in magnitude -0.24392 at $(2,6)$: week 2 (28řF, chill 14) and week 6 (57.8řF, chill 16) sit at opposite ends of the temperature range and pull the fit in opposite directions.

The diagonal, in vector form:

$$h_{ii} = (0.324360,\ 0.445531,\ 0.341802,\ 0.310660,\ 0.184848,\ 0.583833,\ 0.371227,\ 0.437739).$$

The complement $\boldsymbol{I}_n - \boldsymbol{H}$ produces the residuals, $\boldsymbol{e} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}$; its entries are $-h_{ij}$ off the diagonal and $1 - h_{ii}$ on it:

	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8
obs 1	**0.67564**	−0.35620	−0.20895	−0.12715	−0.13487	0.10771	−0.00719	0.05100
obs 2	−0.35620	**0.55447**	−0.12484	−0.02887	−0.19222	0.24392	−0.08457	−0.01169
obs 3	−0.20895	−0.12484	**0.65820**	−0.30944	−0.02537	−0.21131	0.10362	0.11809
obs 4	−0.12715	−0.02887	−0.30944	**0.68934**	−0.02013	−0.30890	0.05733	0.04781
obs 5	−0.13487	−0.19222	−0.02537	−0.02013	**0.81515**	−0.00807	−0.22155	−0.21294
obs 6	0.10771	0.24392	−0.21131	−0.30890	−0.00807	**0.41617**	−0.08071	−0.15882
obs 7	−0.00719	−0.08457	0.10362	0.05733	−0.22155	−0.08071	**0.62877**	−0.39571
obs 8	0.05100	−0.01169	0.11809	0.04781	−0.21294	−0.15882	−0.39571	**0.56226**

The diagonal of $\boldsymbol{I}_n - \boldsymbol{H}$ is the factor $1 - h_{ii}$ in $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii})$. It is smallest for observation 6 ($1 - 0.58383 = 0.41617$): the week with the most unusual predictors has the residual most tightly pinned to zero. That is why raw residuals cannot be compared without standardising.

11.3 Properties of $\boldsymbol{H}$

Proposition 11.3 (The five defining properties)**.** *Let $\boldsymbol{X}$ be $n \times k$ with $\mathrm{rank}(\boldsymbol{X}) = k$ and $\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top$. Then*

$$\begin{aligned}
\boldsymbol{H}^\top &= \boldsymbol{H} && \text{(symmetric)}\\
\boldsymbol{H}\boldsymbol{H} &= \boldsymbol{H} && \text{(idempotent)}\\
\boldsymbol{H}\boldsymbol{X} &= \boldsymbol{X}, \quad \boldsymbol{H}\hat{\boldsymbol{y}} = \hat{\boldsymbol{y}} && (\boldsymbol{H} \text{ fixes its own column space})\\
\mathrm{tr}(\boldsymbol{H}) &= \mathrm{rank}(\boldsymbol{H}) = \mathrm{rank}(\boldsymbol{X}) = k && (\text{so } \mathrm{tr}(\boldsymbol{I}_n - \boldsymbol{H}) = n - k)\\
\boldsymbol{H}(\boldsymbol{I}_n - \boldsymbol{H}) &= \boldsymbol{0} && (\hat{\boldsymbol{y}} \text{ and } \boldsymbol{e} \text{ are orthogonal})
\end{aligned}$$

Proof. Symmetry and idempotence as in Proposition 5.6. $\boldsymbol{H}\boldsymbol{X} = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{X} = \boldsymbol{X}$ and $\boldsymbol{H}\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{y}}$. For an idempotent matrix rank = trace (eigenvalues 0 or 1, the trace counts unit eigenvalues), and $\mathrm{rank}(\boldsymbol{H}) = \mathrm{rank}(\boldsymbol{X})$ because $\boldsymbol{H}$ projects onto $\mathcal{C}(\boldsymbol{X})$. Finally $\boldsymbol{H}(\boldsymbol{I}_n - \boldsymbol{H}) = \boldsymbol{H} - \boldsymbol{H}^2 = \boldsymbol{0}$. □

Proposition 11.4 (Eigenvalues)**.** *$\boldsymbol{H}$ has k eigenvalues equal to 1 and $n - k$ equal to 0. Hence $\sum_{ij} h_{ij}^2 = \mathrm{tr}(\boldsymbol{H}\boldsymbol{H}^\top) = \mathrm{tr}(\boldsymbol{H}) = k$: the "total pull" in the matrix is exactly k.*

Proof. $\boldsymbol{H}$ symmetric gives a real spectral decomposition $\boldsymbol{H} = \boldsymbol{P}\boldsymbol{\Lambda}\boldsymbol{P}^\top$; idempotence forces $\lambda^2 = \lambda$, so $\lambda \in \{0, 1\}$, and $\mathrm{tr}(\boldsymbol{H}) = \sum_j \lambda_j = k$, so exactly k equal 1. Then $\sum_{ij} h_{ij}^2 = \mathrm{tr}(\boldsymbol{H}\boldsymbol{H}^\top) = \mathrm{tr}(\boldsymbol{H}^2) = \mathrm{tr}(\boldsymbol{H}) = k$. □

Numerical verification for the fuel data (max|difference| = max|hand − expected|, all floating-point round-off):

```
(a) symmetry       max |H' - H|              1.554e-15   MATCH
(b) idempotence    max |H H - H|             1.527e-15   MATCH
(c) H X = X        max |H X - X|             3.908e-13   MATCH
(d) I - H is also a projection:
    max |(I-H)' - (I-H)|  1.554e-15   max |(I-H)(I-H) - (I-H)|  1.582e-15   max |H (I-H)|  1.588e-15
(e) tr(H) = 3 = k,  tr(I-H) = 5 = n-k;  rank(H) from qr(): 3,  rank(X): 3
(f) eigenvalues of H:  1 1 1 0 0 0 0 0   (sum = 3 = tr(H) = k)
(g) h_ii = sum_j h_ij^2  for every i
```

11.4 Leverage, from the definition

$h_{ii} = [\boldsymbol{H}]_{ii} = \boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_i$; bounds $0 \le h_{ii} \le 1$, $\sum_i h_{ii} = k$, $\bar{h} = k/n$, flag $h_{ii} > c\,k/n$ with $c \in [2, 4]$ (Proposition 8.2).

obs	1	2	3	4	5	6	7	8	
Temp	28.0	28.0	32.5	39.0	45.9	57.8	58.1	62.5	
Chill	18	14	24	22	8	16	1	0	
h_{ii}	0.3244	0.4455	0.3418	0.3107	0.1848	0.5838	0.3712	0.4377	$\sum = 3$
% of total	10.81	14.85	11.39	10.36	6.16	19.46	12.37	14.59	

$k/n = 3/8 = 0.375$, cut-offs $2k/n = 0.75$ and $3k/n = 1.125$. The largest leverage is observation 6 at 0.5838, below 0.75, so no observation is formally high-leverage; observation 6 also has the smallest $1 - h_{ii}$ and therefore the smallest residual variance. Observation 4, the influential outlier, has $h_{44} = 0.3107$, *below* the average: it is unusual in y, not in $\boldsymbol{x}$.

Common error 11.5. Leverage does not depend on $\boldsymbol{y}$. Changing the fuel values changes $\hat{\boldsymbol{\beta}}$, the residuals, the ANOVA table and every diagnostic except $\boldsymbol{H}$ and h_{ii}. Unusual *in its predictors*: look at h_{ii}; unusual *in its response*: look at the residual. Cook's distance combines the two.

11.5 The residual covariance matrix $\sigma^2(\boldsymbol{I}_n - \boldsymbol{H})$

By Proposition 8.3, $\boldsymbol{e} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{\varepsilon}$, $\mathbb{E}[\boldsymbol{e}] = \boldsymbol{0}$, $\mathrm{Var}(\boldsymbol{e}) = \sigma^2(\boldsymbol{I}_n - \boldsymbol{H})$, so $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii}) \le \sigma^2$ and $\mathrm{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$. The matrix $\boldsymbol{I}_n - \boldsymbol{H}$ displayed above *is* the residual correlation structure up to σ^2. Two consequences:

- **Even with independent errors, the residuals are not independent.** The off-diagonal entries $-\sigma^2 h_{ij}$ are generally non-zero; any argument treating $e_1, \dots, e_n$ as independent is wrong.
- **The residuals are shrunk, unevenly.** $\sum_i \mathrm{Var}(e_i) = \sigma^2 \sum_i (1 - h_{ii}) = \sigma^2(n - k)$, so the average residual variance is $\sigma^2(n-k)/n < \sigma^2$; shrinkage is worst where leverage is highest.

11.6 Standard errors and inference from $(\boldsymbol{X}^\top \boldsymbol{X})^{-1}$

$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{X}^\top \boldsymbol{X})^{-1}$, estimated with $\mathrm{MSE} = \mathrm{RSS}/(n-k) = s^2$. For the fuel example

$$s^2(\boldsymbol{X}^\top \boldsymbol{X})^{-1} = \begin{pmatrix} 0.732219 & -0.011579 & -0.016016 \\ -0.011579 & 0.000198 & 0.000222 \\ -0.016016 & 0.000222 & 0.000484 \end{pmatrix},$$

whose diagonal entries are 0.855698^2, 0.0140774^2, 0.0220025^2, the squared standard errors. The off-diagonals are covariances: $\mathrm{Cov}(\hat{\beta}_1, \hat{\beta}_2) = 0.000222$, giving

$$\mathrm{corr}(\hat{\beta}_1, \hat{\beta}_2) = \frac{0.000222}{0.0140774 \times 0.0220025} = 0.718,$$

$$\mathrm{corr}(\hat{\beta}_0, \hat{\beta}_1) = \frac{-0.011579}{0.855698 \times 0.0140774} = -0.961.$$

(The slope estimates are positively correlated because Temp and Chill themselves are negatively correlated, $\mathrm{corr}(\text{Temp}, \text{Chill}) = -0.718$.)

```
MSE = s^2 = RSS/(n-k) = 0.1347464     s = 0.3670782
               estimate         se          t      p_value       CI_low     CI_high
(Intercept) 13.10873722  0.85569805 15.31934914 0.0000215019 10.90909535 15.30837909
Temp        -0.09001387  0.01407736 -6.39422935 0.0013861048 -0.12620088 -0.05382686
Chill        0.08249497  0.02200255  3.74933734 0.0133028057  0.02593562  0.13905432
t critical value t_{5,0.975} = 2.5705818
CHECK against lm()/summary()/confint(): MSE, standard errors, t values,
estimated covariance matrix  -- all max|difference| = 0.000e+00   MATCH
```

11.7 Standardised and studentised residuals: closed form against refitting

The delete-one closed forms (Proposition 8.5) and an actual refit n times agree to 7.7×10^{-14}:

```
obs          e     h_ii   1-h_ii      r_std  s_delete     r_stud
  1  0.3267417 0.324360 0.675640  1.0828999  0.359068  1.1070583
  2 -0.0432784 0.445531 0.554469 -0.1583341  0.409383 -0.1419747
  3  0.2368342 0.341802 0.658198  0.7952570  0.383569  0.7610579
  4 -0.6130857 0.310660 0.689340 -2.0116193  0.179210 -4.1204154
  5 -0.2370603 0.184848 0.815152 -0.7152881  0.388839 -0.6752537
  6  0.2741450 0.583833 0.416167  1.1576788  0.351115  1.2102933
  7  0.0385738 0.371227 0.628773  0.1325215  0.409691  0.1187395
  8  0.0171298 0.437739 0.562261  0.0622334  0.410253  0.0556849
CHECK against lm(): standardised residuals max|diff| = 3.102e-13   MATCH
                    studentised residuals  max|diff| = 2.153e-12   MATCH
```

$s_{(4)} = 0.17921$ is much smaller than $s = 0.36708$ because observation 4 has the largest residual: leaving it out shrinks the estimated error variance. That is exactly why its standardised residual is -2.01 while its studentised residual is -4.12: the whole discrepancy is the factor $s/s_{(4)} = 0.3670782/0.1792105 = 2.048$; the point inflates its own yardstick.

11.8 Cook's distance: closed form and refitting

```
obs   h_ii   r_stud   D_closed     D_refit
  1 0.3244   1.1071 0.18765834  0.18765834
  2 0.4455  -0.1420 0.00671472  0.00671472
  3 0.3418   0.7611 0.10947400  0.10947400
  4 0.3107  -4.1204 0.60788586  0.60788586
  5 0.1848  -0.6753 0.03867384  0.03867384
  6 0.5838   1.2103 0.62672445  0.62672445
  7 0.3712   0.1187 0.00345618  0.00345618
  8 0.4377   0.0557 0.00100509  0.00100509
D (closed form) vs D (definition, refitting) max|difference| = 1.2e-16   MATCH
cut-off 4/(n-k) = 0.8 ;  D above the cut-off: none (obs 4 has D = 0.6079)
CHECK against lm(): Cook's distance max|difference| = 3.664e-15   MATCH
```

Observations 4 and 6 dominate, for completely different reasons:

obs	h_{ii}	r_i^*	why (with Cook's D)
4	0.3107 (low)	-4.1204 (large)	unusual in y: an outlier whose predictors are ordinary. $D_4 = 0.6079$
6	0.5838 (highest)	1.2103 (moderate)	unusual in $\boldsymbol{x}$: a high-leverage week with an ordinary residual. $D_6 = 0.6267$

So the largest Cook's distance here comes from the high-leverage point, not from the outlier: a corrective to the reflex that "D finds outliers". D_i mixes the two: r_i^2 measures the surprise and $h_{ii}/(1-h_{ii})^2$ the leverage with which that surprise is transmitted to the whole fitted surface.

11.9 The full influence table

The last three columns are $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_i e_i/(1-h_{ii})$, the change in each coefficient when the observation is deleted (verified against actual refits):

```
obs     h_ii      r_std     r_stud          D     dbeta0     dbeta1     dbeta2
  1 0.324360  1.0828999  1.1070583 0.18765834  0.429726  -0.007270  -0.003850
  2 0.445531 -0.1583341 -0.1419747 0.00671472 -0.106467   0.001689   0.001743
  3 0.341802  0.7952570  0.7610579 0.10947400 -0.076007   0.000536   0.007565
  4 0.310660 -2.0116193 -4.1204154 0.60788586  0.473209  -0.006891  -0.021853
  5 0.184848 -0.7152881 -0.6752537 0.03867384 -0.156754   0.001517   0.004169
  6 0.583833  1.1576788  1.2102933 0.62672445 -0.944901   0.016792   0.022431
  7 0.371227  0.1325215  0.1187395 0.00345618  0.019795   0.000072  -0.001187
  8 0.437739  0.0622334  0.0556849 0.00100509  0.001932   0.000182  -0.000477
```

Reading row 4: deleting week 4 lowers the intercept by 0.473 (to 12.636), raises the temperature slope by 0.0069 (from -0.0900 to -0.0831) and raises the chill slope by 0.0219 (to 0.1043). Deleting week 6 moves the coefficients even more (intercept to 14.054, slopes to -0.1068 and 0.0601): one is a y-outlier and one an x-outlier, and both matter.

11.10 Everything else, from the same matrices

Proposition 11.6 (Sums of squares as quadratic forms)**.**

$$\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y}, \quad \boldsymbol{e} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}, \quad \mathrm{RegSS} = \boldsymbol{y}^\top\big(\boldsymbol{H} - \tfrac{1}{n}\mathbf{1}\mathbf{1}^\top\big)\boldsymbol{y}, \quad \mathrm{RSS} = \boldsymbol{y}^\top(\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}.$$

Under $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\boldsymbol{I}_n)$*,* $\mathrm{RSS}/\sigma^2 \sim \chi^2_{n-k}$*; under* $H_0 : \beta_1 = \cdots = \beta_p = 0$*,* $\mathrm{RegSS}/\sigma^2 \sim \chi^2_p$ *independently of RSS, hence* $F = \mathrm{MSR}/\mathrm{MSE} \sim F_{p,n-k}$*.*

Why the ranks are what they are. $\boldsymbol{I}_n - \boldsymbol{H}$ is symmetric idempotent with trace $n-k$, so $\mathrm{RSS}/\sigma^2 = \boldsymbol{z}^\top(\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{z} \sim \chi^2_{n-k}$ for $\boldsymbol{z} = \boldsymbol{\varepsilon}/\sigma$. Under H_0 the mean vector is $\beta_0\mathbf{1}$ and the centring matrix $\boldsymbol{H} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ is symmetric idempotent with trace $k-1=p$; independence follows from $(\boldsymbol{H} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top)(\boldsymbol{I}_n - \boldsymbol{H}) = \mathbf{0}$. □

```
TSS    = sum (y_i - ybar)^2      = 25.54875
Reg SS = sum (y.hat_i - ybar)^2  = 24.87501798
RSS    = sum e_i^2               = 0.67373202        Reg SS + RSS = 25.54875 = TSS
Reg SS: definition vs beta'X'y form     max|difference| = 3.553e-15   MATCH
Reg SS: definition vs y.hat'y.hat form  max|difference| = 3.553e-15   MATCH
ANOVA:  Regression 24.875018 df 2 MS 12.437509  F 92.3031  p 0.00011293
        Error       0.673732 df 5 MS  0.134746
        Total      25.548750 df 7
partial F = (2.5679337 - 0.673732)/1 / 0.1347464 = 14.057530 ;  t(Chill)^2 = 14.057530   <-- t^2 = F
y.hat0 = 11.233271 ; x0'(X'X)^-1 x0 = 0.57458892
95% CI mean response [10.518003, 11.948539] ; 95% PI new obs [10.049211, 12.417331]
```

11.11 A larger case: Duncan prestige data, $n = 45$, $k = 5$

Small n is not what makes the algebra work. Repeat everything for

$$\texttt{prestige} \sim \texttt{income} + \texttt{education} + \texttt{type},$$

where `type` has three levels (`bc`, `prof`, `wc`). The design matrix is built by hand: intercept, two numeric columns, two dummies for `prof` and `wc` (`bc` baseline), so $k = 5$ and $\boldsymbol{H}$ is 45×45.

```
levels of 'type': bc prof wc  -> baseline = bc, dummies for prof and wc; X2 is 45 x 5
beta.hat: (Intercept) -0.185028  income 0.597546  education 0.345319
          typeprof 16.657513  typewc -14.661133
MSE = 94.948869   s = 9.744171
leverage: min = 0.048603616  mean = 0.11111111 = k/n  max = 0.36635189 ;  sum(h_ii) = 5 = k
h_ii > 2k/n = 0.2222 : 9, 16, 17, 19, 21, 27        h_ii > 3k/n = 0.3333 : 16
|r*| > 2 : 6, 28                                    D_i > 4/(n-k) = 0.1 : 6, 17, 19, 21, 27
CHECK against lm(): leverage 2.220e-16, standardised 2.787e-14, studentised 2.853e-14,
                    Cook's distance 3.664e-15   -- all MATCH
```

The mean leverage is exactly $k/n = 5/45 = 0.1111$ and $\sum_i h_{ii} = 5 = k$, as Proposition 8.2 requires, even though two columns are 0/1 dummies rather than measurements. Only observation 16 (`conductor`) exceeds $3k/n$.

What the off-diagonal entries say. The largest off-diagonal entries of $\boldsymbol{H}$:

i	occupation	j	occupation	h_{ij}
19	store.manager	17	contractor	0.22234
23	insurance.agent	9	reporter	0.19737
21	bookkeeper	24	store.clerk	0.18906
6	minister	27	RR.engineer	-0.18851
16	conductor	22	mail.carrier	0.18810
21	bookkeeper	9	reporter	0.18504

Positive entries pair occupations with similar predictor profiles (store manager and contractor, bookkeeper and store clerk, conductor and mail carrier); the negative entry pairs a minister with a railway engineer, at opposite ends of the income/education space. This is the mechanism behind the leverage warning: a very unusual observation does not merely fail to help the fit at the other end of the data, it actively pulls it the wrong way.

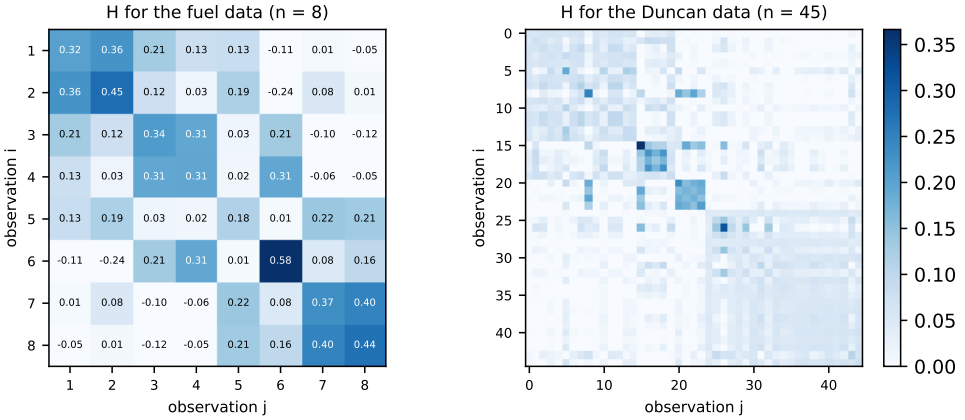


Figure 2: Hat-matrix heat maps. Left: all 64 entries of $\boldsymbol{H}$ for the fuel data. Right: the 45×45 hat matrix for the Duncan data. The diagonal is the darkest cell in each row and the off-diagonal structure fades with distance in predictor space. R: base `image()` with `hcl.colors(64, "Blues 3", rev = TRUE)`.

11.12 The R code: no packages

Each block is three or four lines because $\boldsymbol{H}$ does all the work; `hatvalues`, `rstandard`, `rstudent`, `cooks.distance`, `influence`, `vcov`, `confint`, `predict`, `anova`, `summary` appear only in CHECK blocks as an independent audit. Build first, check second.

```
## --- the design matrix, built by hand (no model.matrix) ---------------------
X <- cbind("(Intercept)" = rep(1, n), Temp = Temp, Chill = Chill)
y <- matrix(Fuel, ncol = 1)
XtX    <- t(X) %*% X
XtXinv <- solve(XtX)                    # (X'X)^-1
beta   <- XtXinv %*% t(X) %*% y         # least squares estimate
## --- THE HAT MATRIX, ALL OF IT ----------------------------------------------
H <- X %*% XtXinv %*% t(X)
print(round(H, 5))                      # all n x n entries
h <- diag(H)                            # leverage
## --- fitted values and residuals --------------------------------------------
yhat <- H %*% y
e    <- y - yhat                        # same as (diag(n) - H) %*% y
## --- leverage, row by row, straight from the definition ---------------------
h_rowwise <- sapply(1:n, function(i)
  as.numeric(t(X[i, , drop = FALSE]) %*% XtXinv %*% X[i, , drop = FALSE]))
## --- residual covariance structure ------------------------------------------
ImH <- diag(n) - H                      # Var(e) = sigma^2 (I - H)
var_e_factor <- diag(ImH)               # the 1 - h_ii
## --- MSE, standard errors, t tests, confidence intervals --------------------
k   <- ncol(X)
MSE <- as.numeric(t(e) %*% e) / (n - k)
Vb  <- MSE * XtXinv                     # estimated Var(beta.hat)
se  <- sqrt(diag(Vb));  tv <- as.numeric(beta) / se;  tc <- qt(0.975, n - k)
CI  <- cbind(as.numeric(beta) - tc * se, as.numeric(beta) + tc * se)
## --- standardised residual --------------------------------------------------
rstd <- as.numeric(e) / (sqrt(MSE) * sqrt(1 - h))
## --- studentised residual: BY REFITTING n times -----------------------------
rstu_refit <- sapply(1:n, function(i) {
  Xi <- X[-i, , drop = FALSE]; yi <- matrix(y[-i, ], ncol = 1)
  bi <- solve(t(Xi) %*% Xi) %*% t(Xi) %*% yi       # refit without obs i
  ei <- yi - Xi %*% bi
  s_i <- sqrt(as.numeric(t(ei) %*% ei) / (n - k - 1))
  e[i] / (s_i * sqrt(1 - h[i]))
})
## --- ... and the closed-form delete-one version -----------------------------
s_i  <- sqrt((as.numeric(t(e) %*% e) - as.numeric(e)^2 / (1 - h)) / (n - k - 1))
rstu <- as.numeric(e) / (s_i * sqrt(1 - h))
## --- Cook's distance: definition (refit) and closed form --------------------
D_def <- sapply(1:n, function(i) {
  Xi <- X[-i, , drop = FALSE]; yi <- matrix(y[-i, ], ncol = 1)
  bi <- solve(t(Xi) %*% Xi) %*% t(Xi) %*% yi
  sum((yhat - X %*% bi)^2) / (k * MSE)
})
D <- (as.numeric(e)^2 / (k * MSE)) * h / (1 - h)^2
## --- delete-one change in the coefficients ----------------------------------
dbeta <- sapply(1:n, function(i)
  (XtXinv %*% matrix(X[i, ], ncol = 1) * as.numeric(e[i])) / (1 - h[i]))
## --- ANOVA, global F, partial F, prediction ---------------------------------
ybar  <- mean(y)
TSS   <- as.numeric(t(y - ybar) %*% (y - ybar))
RegSS <- as.numeric(t(yhat - ybar) %*% (yhat - ybar))
RSS   <- as.numeric(t(e) %*% e)
Fstat <- (RegSS / (k - 1)) / (RSS / (n - k))
x0 <- matrix(c(1, 30, 10), ncol = 1)
se_mean <- sqrt(MSE * as.numeric(t(x0) %*% XtXinv %*% x0))
se_pred <- sqrt(MSE * (1 + as.numeric(t(x0) %*% XtXinv %*% x0)))
## --- heat map, base graphics ------------------------------------------------
pal <- hcl.colors(64, "Blues 3", rev = TRUE)          # dark = large
image(1:n, 1:n, t(H[n:1, ]), col = pal, zlim = c(0, max(abs(H))))
```

11.13 Checklist of the formulas used

Quantity	Formula
Hat matrix	$\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top$
Fitted values	$\hat{\boldsymbol{y}} = \boldsymbol{H}\boldsymbol{y}$
Residuals	$\boldsymbol{e} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}$
Leverage	$h_{ii} = [\boldsymbol{H}]_{ii} = \boldsymbol{x}_i^\top(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{x}_i = \sum_j h_{ij}^2$
Leverage bounds	$0 \le h_{ii} \le 1$, $\sum_i h_{ii} = k$, $\bar{h} = k/n$
Residual covariance	$\mathrm{Var}(\boldsymbol{e}) = \sigma^2(\boldsymbol{I}_n - \boldsymbol{H})$, so $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii})$
Coefficient covariance	$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(\boldsymbol{X}^\top\boldsymbol{X})^{-1}$, estimated by $s^2(\boldsymbol{X}^\top\boldsymbol{X})^{-1}$
Standard error	$\widehat{\mathrm{se}}(\hat{\beta}_j) = \sqrt{\mathrm{MSE}\,[(\boldsymbol{X}^\top\boldsymbol{X})^{-1}]_{jj}}$
Standardised residual	$r_i = e_i / (s\sqrt{1 - h_{ii}})$
Studentised residual	$r_i^* = e_i / (s_{(i)}\sqrt{1 - h_{ii}})$, $s_{(i)}^2 = \dfrac{\mathrm{RSS} - e_i^2/(1 - h_{ii})}{n - k - 1}$
Delete-one fitted values	$\hat{\boldsymbol{y}}_{(i)} = \hat{\boldsymbol{y}} - \boldsymbol{h}_i e_i/(1 - h_{ii})$
Delete-one coefficients	$\hat{\boldsymbol{\beta}}_{(i)} = \hat{\boldsymbol{\beta}} - (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{x}_i e_i/(1 - h_{ii})$
Cook's distance	$D_i = \dfrac{e_i^2}{k\,s^2} \cdot \dfrac{h_{ii}}{(1 - h_{ii})^2}$, cut-off $4/(n - k)$
Sums of squares	$\mathrm{RSS} = \boldsymbol{y}^\top(\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}$, $\mathrm{RegSS} = \boldsymbol{y}^\top(\boldsymbol{H} - \frac{1}{n}\boldsymbol{1}\boldsymbol{1}^\top)\boldsymbol{y}$
Total pull	$\sum_{ij} h_{ij}^2 = \mathrm{tr}(\boldsymbol{H}) = k$

12 Categorical predictors: a change of coordinates

← **My thought**
The theorem is easier to remember when its hypothesis feels necessary.

A categorical predictor with r levels needs $r-1$ independent columns when the model has an intercept. Including all r indicators plus the intercept makes $\boldsymbol{X}$ rank deficient because the indicators sum to the intercept column. With *treatment coding* the omitted level is a reference group and each coefficient is a difference from its mean. With *effect coding* coefficients are deviations under a sum-to-zero constraint. Coefficients and their interpretation change with the coding; the fitted group means need not.

Education example with three levels: no high school, completed high school, post-secondary. Effect-coded rows for $(1, E, F)$ are $(1, 1, 0)$, $(1, 0, 1)$ and $(1, -1, -1)$. Treatment coding with post-secondary as reference uses $(1, D_N, D_C)$ with rows $(1, 1, 0)$, $(1, 0, 1)$ and $(1, 0, 0)$. For the same three observations $\boldsymbol{X}_{\text{effect}} = \boldsymbol{X}_{\text{dummy}}\boldsymbol{T}$ with

$$\boldsymbol{T} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \qquad \det(\boldsymbol{T}) = 3 \neq 0.$$

Thus $\operatorname{col}(\boldsymbol{X}_{\text{effect}}) = \operatorname{col}(\boldsymbol{X}_{\text{dummy}})$. More generally any invertible column change $\boldsymbol{X}_2 = \boldsymbol{X}_1\boldsymbol{T}$ gives $\boldsymbol{\beta}_2 = \boldsymbol{T}^{-1}\boldsymbol{\beta}_1$, the same fitted values and the same projection $\boldsymbol{H}_2 = \boldsymbol{H}_1$. This is *matrix reparameterization*: an individual coefficient may look different, but a properly translated contrast and its inference describe the same group comparison. With two categorical factors, both parameterizations must represent the same intended main effects and interactions before fits can be compared.

Additive wage example. Education uses (E, F) and number of children uses (G, H), each with three levels. Under effect coding, the mean for post-secondary education and more than two children is $a - b_1 - b_2 - c_1 - c_2$, and its contrast with the balanced average of all nine cell means is $-b_1 - b_2 - c_1 - c_2$. A sample-weighted average is a different target when group sizes are unequal; the reference average must be specified before interpreting a contrast.

13 Model selection criteria

Model choice
I read AIC, BIC and adjusted R^2 as different answers to different questions.

One more variable always helps the training RSS. Enlarging the column space gives the optimizer more directions, so RSS cannot increase. That does not mean the new variable helps prediction: a statement about the *training projection* differs from one about the *distribution of future observations*.

Adjusted R^2 is a degrees-of-freedom correction.

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}, \qquad \boxed{R^2_{\text{adj}} = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}}$$

A ratio of two variance estimates: the denominator loses 1 df for the estimated mean and the numerator loses $p+1$ for the fitted coefficients, the same logic as the unbiased variance estimator.

AIC asks about predictive information loss. For k estimated parameters and maximized log-likelihood $\ell(\hat{\theta})$,

$$\boxed{\text{AIC} \quad = \quad -2\ell(\hat{\theta}) \quad + \quad 2k}$$

The $2k$ compensates approximately for the optimistic evaluation of fitted likelihood on the same observations; in regular large-sample models it con-

nects to expected out-of-sample Kullback–Leibler information loss. The numerical AIC is not an absolute goodness measure: comparisons need a common response, consistent likelihood conventions and the same data.

BIC asks a different question.

$$\text{BIC} \quad = \quad -2\ell(\hat{\theta}) \;+\; k \log n$$

The $k \log n$ penalty arises from a large-sample approximation to integrated likelihood (under suitable priors and regularity). Its target and assumptions differ from AIC, so BIC is not simply a "stricter AIC". It often prefers smaller models for large n, but prediction and evidence for a model are different aims. For $n > e^2$ the BIC penalty per parameter exceeds AIC's; include variance and dispersion parameters consistently when comparing full likelihoods.

Example of conflicting decisions. Adding one coefficient raises the maximized log-likelihood by 3, so -2ℓ changes by -6. AIC changes by $-6+2 = -4$ (favours the larger model). At $n = 1000$, BIC changes by $-6 + \log 1000 \approx +0.91$ (favours the smaller model). Neither contradicts arithmetic; they have different inferential targets.

Next to any model-selection table: adjusted R^2 compares explained variance adjusted for df; AIC approximates a prediction-oriented criterion; BIC approximates an evidence-oriented criterion. All depend on the model class and assumptions.

14 Common errors, exercises and reference facts

14.1 Common student errors to anticipate

1. **Using $n - 2$ instead of $n - p - 1$.** Ask: how many parameters did you estimate? The error df is n minus that number.
2. **Confusing RegSS with the sum of the `anova()` lines.** They agree only because the model has an intercept and the lines are sequential; always check $\text{TSS} = \text{RegSS} + \text{RSS}$.
3. **Reading the `anova()` F column as the global test.** It is one predictor at a time.
4. **Reporting R^2 as evidence of a good model.** With enough predictors $R^2 \to 1$ regardless of relevance; report R^2_{adj} and residual plots.
5. **Testing several slopes with several t-tests.** Multiplicity inflates the type I error rate; use one partial F.
6. **Computing $\text{Var}(e_i) = \sigma^2$.** Wrong: $\text{Var}(e_i) = \sigma^2(1 - h_{ii})$ by Proposition 8.3, which is why standardisation is needed.
7. **Deleting an outlier because it is inconvenient.** Diagnose first; report the fit both with and without it.

14.2 Exercises

1. **Check the degrees of freedom.** For the fuel data, state the df for TSS, RegSS, RSS and verify that they add up. What would they be for $y \sim$ Temp only? For $y \sim$ Temp + Chill + Temp $\times$ Chill?
2. **Verify the identity.** Using the residuals of Section 9, confirm numerically that $\sum_i e_i = 0$ and $\sum_i \hat{y}_i e_i = 0$, and explain which normal equation each identity comes from.
3. **Reverse the order.** Refit as `lm(Fuel ~ Chill + Temp)`. Report `anova()` for both orders and explain why the two lines swap but RegSS, the global F, the coefficients and the fitted values do not change.

4. **Prove the $t^2 = F$ identity.** Show that for a single omitted predictor $\mathrm{RSS(R)} - \mathrm{RSS(F)} = \hat\beta_j^2 / \widehat{\mathrm{se}}(\hat\beta_j)^2 \cdot \mathrm{MSE}$, and deduce Corollary 7.3.
5. **Derive the prediction interval.** For a new observation with predictor vector $\boldsymbol{x}_0$, show $\mathrm{Var}(\hat y_0 - \varepsilon_0) = \sigma^2(1 + \boldsymbol{x}_0^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_0)$ and write down the $100(1-\alpha)\%$ prediction interval. Why is $\hat y_0$ independent of ε_0?
6. **A high-leverage case.** Add a ninth observation with temperature 75řF, chill index 0, fuel use 7.4. Recompute $\hat{\boldsymbol{\beta}}$, the leverages, the ANOVA table and the Cook distances. (Answer: its own leverage $h_{99} = 0.4497$ is the largest in the data set, and *every* standard error grows, $\widehat{\mathrm{se}}(\hat\beta_1)$ from 0.01408 to 0.01557, so all confidence intervals widen. The coefficients move only moderately, roughly 15%, because the new point lies fairly close to the fitted surface. Leverage is not the same as influence.)
7. **Collinearity.** Compute the correlation between temperature and chill index (-0.718) and the $(\boldsymbol{X}^\top \boldsymbol{X})^{-1}$ diagonal entries. Replace the chill index by $2\times$chill and recompute. Which numbers change, and which do not? Relate the answer to VIF.
8. **No intercept.** Refit without an intercept and confirm $\mathrm{RSS} = 32.296350 > \mathrm{TSS} = 25.548750$. Explain why this does not contradict $\mathrm{TSS} = \mathrm{RegSS} + \mathrm{RSS}$.

14.3 Matrix-calculus and trace facts used

1. $\dfrac{\partial(\boldsymbol{a}^\top \boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \boldsymbol{a}$, and for symmetric $\boldsymbol{A}$, $\dfrac{\partial(\boldsymbol{\beta}^\top \boldsymbol{A}\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = 2\boldsymbol{A}\boldsymbol{\beta}$.
2. For symmetric idempotent $\boldsymbol{A}$: all eigenvalues are 0 or 1; $\mathrm{rank}(\boldsymbol{A}) = \mathrm{tr}(\boldsymbol{A})$; $\boldsymbol{A}(\boldsymbol{I}_n - \boldsymbol{A}) = \boldsymbol{0}$.
3. For $\boldsymbol{z}$ with $\mathbb{E}[\boldsymbol{z}] = \boldsymbol{\mu}$, $\mathrm{Var}(\boldsymbol{z}) = \boldsymbol{\Sigma}$ and fixed $\boldsymbol{M}$: $\mathbb{E}[\boldsymbol{z}^\top \boldsymbol{M} \boldsymbol{z}] = \mathrm{tr}(\boldsymbol{M}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \boldsymbol{M} \boldsymbol{\mu}$.
4. $\mathrm{tr}(\boldsymbol{A}\boldsymbol{B}) = \mathrm{tr}(\boldsymbol{B}\boldsymbol{A})$, so $\mathrm{tr}(\boldsymbol{A}\sigma^2 \boldsymbol{I}_n \boldsymbol{A}^\top) = \sigma^2 \,\mathrm{tr}(\boldsymbol{A}\boldsymbol{A}^\top)$, the sum of squared entries of $\boldsymbol{A}$.
5. If $\boldsymbol{z} \sim N_n(\boldsymbol{0}, \boldsymbol{I}_n)$ and $\boldsymbol{A}, \boldsymbol{B}$ are symmetric idempotent with $\boldsymbol{A}\boldsymbol{B} = \boldsymbol{0}$, then $\boldsymbol{z}^\top \boldsymbol{A} \boldsymbol{z} \sim \chi^2_{\mathrm{rank}(\boldsymbol{A})}$ independently of $\boldsymbol{z}^\top \boldsymbol{B} \boldsymbol{z}$ (Cochran; Lemma 6.8, the engine of Theorem 6.9).
6. If $U \sim \chi^2_r$, $V \sim \chi^2_s$, $U \perp\!\!\!\perp V$, then $\dfrac{U/r}{V/s} \sim F_{r,s}$; if additionally $Z \sim N(0,1)$ with $Z \perp\!\!\!\perp V$, then $\dfrac{Z}{\sqrt{V/s}} \sim t_s$.

14.4 Notation summary

$\boldsymbol{y}, \boldsymbol{X}, \boldsymbol{\beta}, \boldsymbol{\varepsilon}$	response vector, design matrix, parameter vector, error vector
n, p, k	sample size, number of predictors, number of parameters $k = p + 1$
$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$	least squares estimator
$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{H}\boldsymbol{y}$	fitted values; $\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top$ (hat matrix)
$\boldsymbol{e} = \boldsymbol{y} - \hat{\boldsymbol{y}} = (\boldsymbol{I}_n - \boldsymbol{H})\boldsymbol{y}$	residual vector
$h_{ii} = \boldsymbol{x}_i^\top (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{x}_i$	leverage of observation i
TSS, RegSS, RSS	total, regression and residual sums of squares
$\mathrm{MSR} = \mathrm{RegSS}/p$, $\mathrm{MSE} = \mathrm{RSS}/(n-p-1) = s^2$	mean squares; $\mathbb{E}[\mathrm{MSE}] = \sigma^2$
$F = \mathrm{MSR}\,/\,\mathrm{MSE}$	global test statistic, $\sim F_{p,n-p-1}$ under H_0
$R^2 = \mathrm{RegSS}\,/\,\mathrm{TSS}$	coefficient of determination (never decreases as p grows)
r_i, r_i^*	standardised and studentised residuals
D_i	Cook's distance, cut-off $4/(n-p-1)$
$\mathrm{VIF}_j = 1/(1-R_j^2)$	variance inflation factor of predictor j

Part III. Generalized linear models

15 Likelihood and maximum likelihood estimation

⟵ **Motivation**
Inference uses data to distinguish plausible parameters; assumptions still matter.

For independent observations with density or mass $f(y_i;\theta)$,

$$L(\theta;y)=\prod_i f(y_i;\theta), \qquad \ell(\theta;y)=\sum_i \log f(y_i;\theta).$$

The data are fixed in this optimization and θ varies; maximizing L or ℓ gives the same estimate. A zero derivative identifies a candidate stationary point, not automatically a global maximum or a boundary solution.

Examples. For Bernoulli responses the MLE of the common success probability is the sample proportion. For independent exponential observations with rate λ, $\hat\lambda = 1/\bar y$. Under a normal model $\hat\mu=\bar y$ and $\hat\sigma^2_{\text{MLE}} = n^{-1}\sum_i (y_i-\bar y)^2$, whereas the unbiased sample variance uses $n-1$. The same distinction appears in normal regression: the MLE of σ^2 divides RSS by n, the residual mean square divides by the residual degrees of freedom.

16 Generalized linear models and the exponential family

⟵ **Proof idea**
Check the initial and boundary conditions after finding a general solution.

A GLM keeps a linear *predictor* $\eta_i = \boldsymbol{x}_i^\top\boldsymbol{\beta}$ while letting the response distribution and the mean-to-predictor relationship change. Three parts:

1. **Random component:** conditionally independent Y_i from a specified exponential dispersion family $f(y_i;\theta,\phi)$.
2. **Systematic component:** $\eta_i = \boldsymbol{x}_i^\top\boldsymbol{\beta}$, with $\boldsymbol{y}=(y_1,\dots,y_n)^\top$, $\boldsymbol{X}=(\boldsymbol{x}_1,\dots,\boldsymbol{x}_n)^\top$, $\boldsymbol{\beta}=(\beta_0,\beta_1,\dots,\beta_p)^\top$.
3. **Link:** $g(\mu_i)=\eta_i$, where $\mu_i = E(Y_i\mid \boldsymbol{x}_i)$ is related to the canonical parameter by $\mu_i = b'(\theta_i)$.

Why normal linear regression is not always sensible. For a count, a real-valued linear prediction may be negative; for a probability, an identity-link fit may leave $[0,1]$; if the variance grows with the mean, constant variance is unsuitable. A GLM is a *two-part modeling choice*: a response distribution and a link from its conditional mean to the linear predictor. The equations $\eta_i=\boldsymbol{x}_i^\top\boldsymbol{\beta}$, $g(\mu_i)=\eta_i$ do not determine the variance; that comes from the family.

Positive outcomes. Fitting a line on the original scale can predict negative means. Regressing $\log Y$ changes the response and targets $E(\log Y\mid x)$, which is not in general $\log E(Y\mid x)$. A GLM with a log link specifies $\log E(Y\mid x)=\boldsymbol{x}^\top\boldsymbol{\beta}$ directly with an appropriate family and variance.

16.1 Exponential dispersion family: mean and variance from one formula

$$f(y;\theta,\phi)=\exp\left\{\frac{y\theta-b(\theta)}{a(\phi)}+c(y,\phi)\right\}.$$

Exponential families
I derive the mean and variance from $b(\theta)$ instead of learning the family table by heart.

θ is the natural (canonical) parameter and ϕ controls dispersion. For fixed ϕ,

$$s_\theta(y) = \frac{y - b'(\theta)}{a(\phi)}, \qquad \frac{\partial^2 \ell}{\partial \theta^2} = -\frac{b''(\theta)}{a(\phi)}.$$

The expected score is zero under regularity conditions, and the score variance equals the negative expected curvature, so

$$\mu \;=\; E(Y) \;=\; b'(\theta), \qquad \mathrm{Var}(Y) \;=\; a(\phi)\, b''(\theta) \;=\; a(\phi)\, V(\mu)$$

with *variance function* $V(\mu) = b''(\theta(\mu))$. The link is *canonical* when $g(\mu) = \theta(\mu)$. Mean and variance are linked rather than unrelated assumptions: $a(\phi)$ sets the scale, $V(\mu)$ the shape of the mean–variance relationship.

Response family	Mean	$V(\mu)$	Common link
Gaussian	μ	1	identity
Bernoulli	μ	$\mu(1-\mu)$	logit
Poisson	μ	μ	log
Gamma	μ	μ^2	inverse (up to parameterization)
Inverse Gaussian	μ	μ^3	inverse-squared (up to scale)

Distribution examples to derive. Bernoulli: $\theta = \log[\pi/(1-\pi)]$, $b(\theta) = \log(1+e^\theta)$, $b'(\theta) = \pi$, $b''(\theta) = \pi(1-\pi)$. Poisson: $\theta = \log\mu$, $b(\theta) = e^\theta$, $V(\mu) = \mu$. Normal with known variance: $\theta = \mu = \eta$, $\phi = \sigma^2$, $a(\phi) = \phi$, $b(\theta) = \theta^2/2$, $c(y,\phi) = -\frac{1}{2}\big(\frac{y^2}{\phi} + \ln(2\pi\phi)\big)$, variance function constant. Gamma: $V \propto \mu^2$; inverse Gaussian: $V \propto \mu^3$. The crucial question: **what is a property of the distribution and what is a choice imposed with the link?** The log link is not intrinsic to Poisson probability, nor logit to Bernoulli probability.

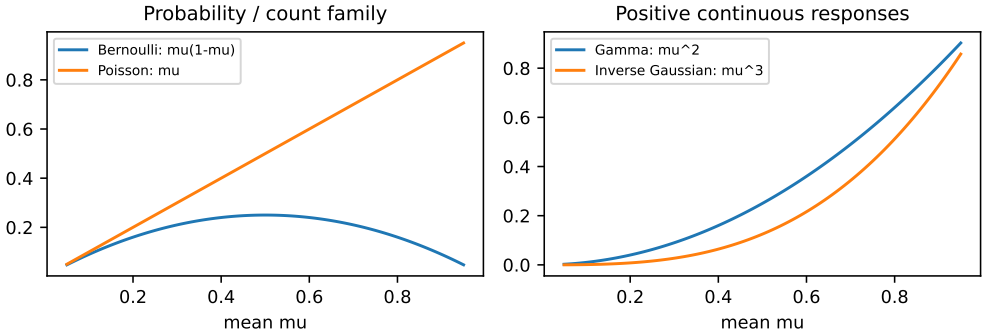


Figure 3: Variance functions for Bernoulli, Poisson, Gamma and inverse Gaussian models. R: `curve()`, `plot()`, `lines()`, or `ggplot2::geom_line()`.

16.2 Links and interpretations

← **Motivation**
Why does this definition matter later? Try an example and a counterexample.

Gaussian response with identity link, $\mu_i = \boldsymbol{x}_i^\top \boldsymbol{\beta}$, recovers normal linear regression. For Poisson counts, $\log \mu_i = \boldsymbol{x}_i^\top \boldsymbol{\beta}$ guarantees $\mu_i = \exp(\boldsymbol{x}_i^\top \boldsymbol{\beta}) > 0$; a one-unit change in a predictor multiplies the mean by e^{β_j}, other predictors fixed. For binary outcomes the logistic model uses

$$\log \frac{\pi_i}{1-\pi_i} = \boldsymbol{x}_i^\top \boldsymbol{\beta}, \qquad \pi_i = \frac{e^{\boldsymbol{x}_i^\top \boldsymbol{\beta}}}{1 + e^{\boldsymbol{x}_i^\top \boldsymbol{\beta}}},$$

so e^{β_j} is a conditional odds ratio for a one-unit change. The probit alternative uses $\Phi^{-1}(\pi_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta}$. Both keep fitted probabilities in $(0,1)$; their coefficient scales differ.

17 The parameter chain: canonical and non-canonical links

Link functions
I draw $\beta \to \eta \to \mu \to \theta$ every time I take derivatives.

The most important diagram in these notes:

$$\boxed{\boldsymbol{\beta}} \xrightarrow{\boldsymbol{x}_i^\top \boldsymbol{\beta}} \boxed{\eta_i} \xrightarrow{g^{-1}} \boxed{\mu_i} \xrightarrow{(b')^{-1}} \boxed{\theta_i}.$$

Back to β. Given a link g,

$$\theta_i = (b')^{-1}(\mu_i) = (b')^{-1}\big(g^{-1}(\boldsymbol{x}_i^\top \boldsymbol{\beta})\big) := h(\boldsymbol{x}_i^\top \boldsymbol{\beta}), \qquad h = (b')^{-1} \circ g^{-1} = (g \circ b')^{-1}.$$

If g is the canonical link function, h is the identity: $\theta_i = \eta_i = \boldsymbol{x}_i^\top \boldsymbol{\beta}$.

The linear predictor η, conditional mean μ and natural parameter θ are normally three different quantities; choosing coefficients, family and link fixes the chain. The canonical link $g = (b')^{-1}$ simplifies score equations; it does not make non-canonical links invalid.

Bernoulli example, $\mu = 0.8$. The natural parameter and logit predictor equal $\log 4 \approx 1.3863$. Under probit the predictor is $\Phi^{-1}(0.8) \approx 0.8416$ although the conditional mean is unchanged. The number 0.8 is neither the coefficient nor the linear predictor; it is the mean.

Derivatives are a chain.

$$\frac{\partial \ell}{\partial \beta_j} = \frac{\partial \ell}{\partial \theta}\frac{\partial \theta}{\partial \mu}\frac{\partial \mu}{\partial \eta}\frac{\partial \eta}{\partial \beta_j}, \qquad \frac{d\mu}{d\theta} = b''(\theta) = V(\mu), \quad \frac{d\eta}{d\mu} = g'(\mu),$$

$$\frac{\partial \ell}{\partial \beta_j} = \sum_i \frac{(y_i - \mu_i)x_{ij}}{a(\phi)V(\mu_i)g'(\mu_i)}$$

For a canonical link $g'(\mu) = 1/V(\mu)$, the denominator reduces to $a(\phi)$ and the score is proportional to $\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{\mu})$, which is why it looks so similar to the normal equations.

Link domains. A log link guarantees a positive mean. An inverse link constrains the admissible predictor, often requiring $\eta > 0$ or $\eta < 0$ depending on sign convention. Even a mathematically valid link must be checked for compatibility with the desired fitted means and predictor values.

18 MLE for GLMs: score, Fisher information and three algorithms

Information & IRLS
The IRLS weights arise from Fisher information, not an arbitrary regression trick.

18.1 Log-likelihood and the normal equations

With independent $Y_i \sim f(y_i; \theta_i, \phi)$,

$$\ell(\boldsymbol{\theta}, \phi; \boldsymbol{Y}) = \sum_{i=1}^n \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + \sum_{i=1}^n c(y_i; \phi) = \ell(\boldsymbol{\beta}),$$

since $\boldsymbol{\beta}$ determines μ_i and in turn θ_i. With the canonical link the expression simplifies to

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \frac{y_i \boldsymbol{x}_i^\top \boldsymbol{\beta} - b(\boldsymbol{x}_i^\top \boldsymbol{\beta})}{a(\phi)} + \sum_{i=1}^n c(y_i; \phi).$$

Assuming the derivatives exist, the normal (likelihood) equations are, by the chain rule,

$$\begin{aligned}\frac{\partial \ell(\boldsymbol{\beta})}{\partial \beta_j} &= \sum_{i=1}^{n} \frac{\partial \ell}{\partial \theta_i}\frac{\partial \theta_i}{\partial \mu_i}\frac{\partial \mu_i}{\partial \eta_i}\frac{\partial \eta_i}{\partial \beta_j} \\ &= \sum_{i=1}^{n} \frac{y_i - b'(\theta_i)}{a(\phi)}\frac{1}{b''(\theta_i)}\frac{\partial \mu_i}{\partial \eta_i}x_{ij} \qquad \Big(\text{recall } \frac{\partial \theta_i}{\partial \mu_i} = \frac{1}{\partial \mu_i/\partial \theta_i}\Big) \\ &= \sum_{i=1}^{n} \frac{y_i - \mu_i}{a(\phi)}\frac{1}{b''(\theta_i)}\frac{x_{ij}}{g'(\mu_i)} = 0, \qquad j = 0, 1, \ldots, p.\end{aligned}$$

Define the **weights** and the **working (adjusted) response**

$$w_i \;=\; \frac{1}{b''(\theta_i)g'(\mu_i)^2} \;=\; \frac{1}{V(\mu_i)g'(\mu_i)^2}, \qquad \xi_i \;=\; \boldsymbol{x}_i^\top\boldsymbol{\beta} + g'(\mu_i)(y_i - \mu_i)$$

Then

$$\frac{\partial \ell(\boldsymbol{\beta})}{\partial \beta_j} = \frac{1}{a(\phi)}\sum_{i=1}^{n}\frac{x_{ij}}{b''(\theta_i)g'(\mu_i)^2}g'(\mu_i)(y_i - \mu_i) = \frac{1}{a(\phi)}\sum_{i=1}^{n} w_i x_{ij}(\xi_i - \boldsymbol{x}_i^\top\boldsymbol{\beta}),$$

and for each i,

$$\mathbb{E}[\xi_i] = \boldsymbol{x}_i^\top\boldsymbol{\beta}, \qquad \operatorname{Var}\{\xi_i\} = \big(g'(\mu_i)\big)^2 V(\mu_i)a(\phi) = w_i^{-1}a(\phi).$$

High variance and a steep link both weaken information about the linear predictor; ξ_i linearizes the link around the current mean.

Score statistics. $U_j := \partial\ell/\partial\beta_j = \frac{1}{a(\phi)}\sum_i w_i x_{ij}(\xi_i - \boldsymbol{x}_i^\top\boldsymbol{\beta})$. The gradient vector (score) $\boldsymbol{U} = (U_0, U_1, \ldots, U_p)^\top$ is

$$\boldsymbol{U} \;=\; \frac{\nabla\ell(\boldsymbol{\beta})}{\nabla\boldsymbol{\beta}} \;=\; \frac{1}{a(\phi)}\sum_{i=1}^{n} w_i\boldsymbol{x}_i(\xi_i - \boldsymbol{x}_i^\top\boldsymbol{\beta}) \;=\; \frac{1}{a(\phi)}\boldsymbol{X}^\top\boldsymbol{W}(\boldsymbol{\xi} - \boldsymbol{X}\boldsymbol{\beta})$$

with $\boldsymbol{X} = (x_{ij})_{n\times(p+1)}$, $\boldsymbol{\xi} = (\xi_1, \ldots, \xi_n)^\top$, $\boldsymbol{W} = \operatorname{diag}(w_1, \ldots, w_n)$. The likelihood equation $\nabla\ell/\nabla\boldsymbol{\beta} = \boldsymbol{0}$ becomes

$$\boldsymbol{X}^\top\boldsymbol{W}(\boldsymbol{\xi} - \boldsymbol{X}\boldsymbol{\beta}) = \boldsymbol{0}, \qquad \text{or} \qquad \boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{\xi}.$$

Example 18.1 (Normal distribution with linear regression)**.** $\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \sim \mathrm{MVN}(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2 I_n)$. The link is $g(\mu) = \mu$, $g'(\mu) = 1$, $V(\mu) = 1$, so

$$w_i = \frac{1}{V(\mu_i)g'(\mu_i)^2} = 1, \qquad \xi_i = \boldsymbol{x}_i^\top\boldsymbol{\beta} + y_i - \mu_i = y_i.$$

The likelihood equation is $\boldsymbol{X}^\top I_n(\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}) = \boldsymbol{0}$, or $\boldsymbol{X}^\top\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}^\top\boldsymbol{Y}$, with an explicit solution when $\boldsymbol{X}^\top\boldsymbol{X}$ is invertible.

Example 18.2 (Poisson distribution with log-linear regression)**.** $y_i \sim \mathrm{Poisson}(\mu_i)$ independent, $\log(\mu_i) = \beta_0 + \beta_1 x_i$. Link $g(\mu) = \log\mu$, $V(\mu) = \mu$. Hence

$$w_i = \mu_i = e^{\beta_0 + \beta_1 x_i}, \qquad \xi_i = \beta_0 + \beta_1 x_i + \mu_i^{-1}(y_i - \mu_i),$$

in matrix form $\boldsymbol{\xi} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{W}^{-1}(\boldsymbol{Y} - \boldsymbol{\mu})$. The normal equation $\boldsymbol{X}^\top\boldsymbol{W}(\boldsymbol{\xi} - \boldsymbol{X}\boldsymbol{\beta}) = \boldsymbol{0}$ reduces to $\boldsymbol{X}^\top(\boldsymbol{Y} - \boldsymbol{\mu}) = \boldsymbol{0}$.

Example 18.3 (Bernoulli with logit link)**.** $W_{ii} = \pi_i(1 - \pi_i)$ and $\xi_i = \eta_i + (y_i - \pi_i)/[\pi_i(1 - \pi_i)]$; the score again reduces to $\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{\pi}) = \boldsymbol{0}$ (canonical link).

18.2 Why iterative algorithms

The normal equation $\boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{\xi}$ cannot in general be solved by writing $\boldsymbol{\beta} = (\boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{\xi}$, because $\boldsymbol{W}$ and $\boldsymbol{\xi}$ depend on $\boldsymbol{\beta}$. Three methods: Newton–Raphson, Fisher scoring, iteratively re-weighted least squares (IRLS).

18.3 The score and Fisher information

The gradient $\boldsymbol{U}(\boldsymbol{\beta};\boldsymbol{Y}) = \nabla\ell/\nabla\boldsymbol{\beta}$ is the *score*; the MLE $\hat{\boldsymbol{\beta}}$ solves $\boldsymbol{U}(\hat{\boldsymbol{\beta}};\boldsymbol{Y}) = \boldsymbol{0}$. Minus the second derivative,

$$\mathcal{J}(\boldsymbol{\beta}) = -\frac{\nabla^2}{\nabla\boldsymbol{\beta}\nabla\boldsymbol{\beta}^\top}\ell(\boldsymbol{\beta};\boldsymbol{Y}),$$

is the *observed information*, measuring how much information is available for estimating $\boldsymbol{\beta}$. The *expected (Fisher) information* is

$$\mathcal{I}(\boldsymbol{\beta}) = \mathbb{E}\Big[-\frac{\partial^2}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top}\ell(\boldsymbol{\beta};\boldsymbol{Y})\Big] = \mathbb{E}[\mathcal{J}(\boldsymbol{\beta})],$$

the average information for the parameters from this model at the specified parameter values.

Lemma 18.4 (An important lemma for iterative methods)**.**

$$\mathbb{E}\Big[\frac{\partial^2\ell}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top}\Big] = -\mathbb{E}\Big[\frac{\partial\ell}{\partial\boldsymbol{\beta}}\frac{\partial\ell}{\partial\boldsymbol{\beta}^\top}\Big],$$

shown in the same manner as the mean–variance property of the exponential dispersion family. It supports

$\mathcal{I}(\boldsymbol{\beta}) \quad = \quad \boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X}$ *(up to the factor* $1/a(\phi)$*, which in most cases is ignored).*

Why $\boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X}$*.* $\boldsymbol{U} = \frac{1}{a(\phi)}\boldsymbol{X}^\top\boldsymbol{W}(\boldsymbol{\xi} - \boldsymbol{X}\boldsymbol{\beta})$ and $\mathrm{Var}(\boldsymbol{\xi}) = a(\phi)\boldsymbol{W}^{-1}$ (diagonal, independent observations), so

$$\mathcal{I} = \mathbb{E}[\boldsymbol{U}\boldsymbol{U}^\top] = \frac{1}{a(\phi)^2}\boldsymbol{X}^\top\boldsymbol{W}\,\mathbb{E}[(\boldsymbol{\xi} - \boldsymbol{X}\boldsymbol{\beta})(\boldsymbol{\xi} - \boldsymbol{X}\boldsymbol{\beta})^\top]\,\boldsymbol{W}\boldsymbol{X}$$

$$= \frac{1}{a(\phi)^2}\boldsymbol{X}^\top\boldsymbol{W}\,a(\phi)\boldsymbol{W}^{-1}\,\boldsymbol{W}\boldsymbol{X} = \frac{1}{a(\phi)}\boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X}. \qquad \square$$

18.4 Newton–Raphson algorithm

Solve $\nabla\ell(\boldsymbol{\beta})/\nabla\boldsymbol{\beta} = \boldsymbol{0}$ by successive linearization. **Step one:** $\boldsymbol{\beta}^{(0)} \to \boldsymbol{\beta}^{(1)}$**.** Take $\boldsymbol{\beta}^{(0)}$ as the starting value and expand about it:

$$\frac{\nabla\ell(\boldsymbol{\beta})}{\nabla\boldsymbol{\beta}}\Big|_{\boldsymbol{\beta}^{(1)}} \approx \frac{\nabla\ell(\boldsymbol{\beta})}{\nabla\boldsymbol{\beta}}\Big|_{\boldsymbol{\beta}^{(0)}} + \frac{\nabla^2\ell}{\nabla\boldsymbol{\beta}\nabla\boldsymbol{\beta}^\top}\Big|_{\boldsymbol{\beta}^{(0)}}(\boldsymbol{\beta}^{(1)} - \boldsymbol{\beta}^{(0)}).$$

Set the left side to $\boldsymbol{0}$ (we seek $\hat{\boldsymbol{\beta}}$ with zero score) and solve

$$\boldsymbol{0} = \frac{\nabla\ell(\boldsymbol{\beta})}{\nabla\boldsymbol{\beta}}\Big|_{\boldsymbol{\beta}^{(0)}} + \frac{\nabla^2\ell}{\nabla\boldsymbol{\beta}\nabla\boldsymbol{\beta}^\top}\Big|_{\boldsymbol{\beta}^{(0)}}(\boldsymbol{\beta}^{(1)} - \boldsymbol{\beta}^{(0)}),$$

giving $\boldsymbol{\beta}^{(1)}$ as a linear function of $\boldsymbol{\beta}^{(0)}$. **Step two:** $\boldsymbol{\beta}^{(1)} \to \boldsymbol{\beta}^{(2)}$ in the same way, and so on:

$\boldsymbol{\beta}^{(k+1)} \quad = \quad \boldsymbol{\beta}^{(k)} \;+\; \mathcal{J}(\boldsymbol{\beta}^{(k)})^{-1}\boldsymbol{U}(\boldsymbol{\beta}^{(k)})$

The process gives $\boldsymbol{\beta}^{(k)} \to \hat{\boldsymbol{\beta}}$. Convergence for GLMs is usually remarkably quick; in practice stop when $\ell(\boldsymbol{\beta}^{(k)})$ and $\ell(\boldsymbol{\beta}^{(k-1)})$ are sufficiently close, often after 4 or 5 iterations. Some extreme data configurations make the log-likelihood have no finite maximum; the algorithm should then report failure to converge, possibly with strangely large estimates and standard errors (complete separation in logistic regression is the standard case).

18.5 Fisher-scoring algorithm

Disadvantages of Newton–Raphson: it is built for a deterministic problem (what about random data?), and the Hessian $\nabla^2\ell/\nabla\boldsymbol{\beta}\nabla\boldsymbol{\beta}^\top$ is complicated to compute. **Idea:** replace the Hessian by its expected value, i.e. update with the Fisher information $\mathcal{I}$:

$$\boldsymbol{\beta}^{(k+1)} \quad = \quad \boldsymbol{\beta}^{(k)} \; + \; \mathcal{I}(\boldsymbol{\beta}^{(k)})^{-1}\boldsymbol{U}(\boldsymbol{\beta}^{(k)})$$

Essentially the same convergence properties as Newton–Raphson, but $\mathcal{I}$ is often easier than $\mathcal{J}$. The inverse covariance matrix of $\hat{\boldsymbol{\beta}}$, $-\mathbb{E}[\nabla^2\ell/\nabla\boldsymbol{\beta}\nabla\boldsymbol{\beta}^\top]$, is estimated by replacing $\boldsymbol{\beta}$ with $\hat{\boldsymbol{\beta}}$.

18.6 Iteratively re-weighted least squares (IRLS)

IRLS is the algorithm for fitting a GLM obtained from Fisher scoring. Recall $\boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{\xi}$ with $w_i = 1/[V(\mu_i)g'(\mu_i)^2]$, $\xi_i = \boldsymbol{x}_i^\top\boldsymbol{\beta} + g'(\mu_i)(y_i - \mu_i)$. Iteration $\boldsymbol{\beta}^{(k)} \to \boldsymbol{\beta}^{(k+1)}$:

1. fix $\boldsymbol{\beta}^{(k)}$ and $\mu_i^{(k)} = g^{-1}(\boldsymbol{x}_i^\top\boldsymbol{\beta}^{(k)})$;
2. compute $\xi_i^{(k)} = \boldsymbol{x}_i^\top\boldsymbol{\beta}^{(k)} + g'(\mu_i^{(k)})(y_i - \mu_i^{(k)})$ and the weights $\boldsymbol{W}^{(k)} = \boldsymbol{W}(\boldsymbol{\beta}^{(k)})$;
3. derive the new estimate from $\boldsymbol{\beta}^{(k)}$, $\boldsymbol{W}^{(k)}$, $\boldsymbol{\xi}^{(k)}$:

$$\boldsymbol{\beta}^{(k+1)} \quad = \quad (\boldsymbol{X}^\top\boldsymbol{W}^{(k)}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{W}^{(k)}\boldsymbol{\xi}^{(k)}$$

Repeat until convergence. It is not one ordinary regression: $\boldsymbol{W}$ and $\boldsymbol{\xi}$ are updated each iteration. For IRLS one only needs $\boldsymbol{X}$, $\boldsymbol{y}$, the link g and the variance function $V(\mu) = b''(\theta)$. A possible starting value is $\boldsymbol{\mu}^{(0)} = \boldsymbol{y}$ (adjusted away from boundary values such as 0 or 1).

Equivalence of the Fisher-scoring step and the IRLS step. $\mathcal{I} = \boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X}/a(\phi)$ and $\boldsymbol{U} = \boldsymbol{X}^\top\boldsymbol{W}(\boldsymbol{\xi} - \boldsymbol{X}\boldsymbol{\beta})/a(\phi)$, so

$$\boldsymbol{\beta}^{(k)} + \mathcal{I}^{-1}\boldsymbol{U} = \boldsymbol{\beta}^{(k)} + (\boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{W}(\boldsymbol{\xi} - \boldsymbol{X}\boldsymbol{\beta}^{(k)}) = (\boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{\xi}.$$

Proposition 18.5. *If the canonical link is used, Fisher scoring is the same as Newton–Raphson.*

Proof. With the canonical link $\theta_i = \eta_i = \boldsymbol{x}_i^\top\boldsymbol{\beta}$, so $\ell(\boldsymbol{\beta}) = \sum_i [y_i\boldsymbol{x}_i^\top\boldsymbol{\beta} - b(\boldsymbol{x}_i^\top\boldsymbol{\beta})]/a(\phi) + \sum_i c(y_i;\phi)$ and

$$\frac{\partial\ell}{\partial\boldsymbol{\beta}} = \frac{1}{a(\phi)}\sum_i \big(y_i - b'(\boldsymbol{x}_i^\top\boldsymbol{\beta})\big)\boldsymbol{x}_i, \qquad \frac{\partial^2\ell}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top} = -\frac{1}{a(\phi)}\sum_i b''(\boldsymbol{x}_i^\top\boldsymbol{\beta})\boldsymbol{x}_i\boldsymbol{x}_i^\top.$$

The Hessian does not involve $\boldsymbol{y}$, so it equals its expectation: $\mathcal{J}(\boldsymbol{\beta}) = \mathcal{I}(\boldsymbol{\beta})$, and the two updates coincide. Moreover $g'(\mu) = 1/b''(\theta)$ gives $w_i = b''(\theta_i) = V(\mu_i)$, so $\mathcal{I} = \boldsymbol{X}^\top\boldsymbol{W}\boldsymbol{X}/a(\phi)$ with $\boldsymbol{W} = \operatorname{diag}(b''(\theta_i))$, consistent with the general formula. For a non-canonical link the Hessian contains extra terms proportional to $(y_i - \mu_i)$, which vanish only in expectation, so $\mathcal{J} \neq \mathcal{I}$ and the two algorithms generally differ. □

Checks before trusting the algorithm. Sufficient rank of the design, fitted means inside the permissible domain, non-singular weights, and a stabilizing likelihood or deviance. Complete separation in logistic regression can make the finite MLE fail to exist while iterates grow steadily. Non-canonical links may need cautious starting values or damping.

19 Deviance: discrepancy from a saturated fit

Deviance

I compare likelihood against a saturated model and handle boundary probabilities with limits.

19.1 Saturated models

The deviance measures the discrepancy of the fitted GLM from a *perfect model* for the sample $(\boldsymbol{X}, \boldsymbol{y})$. Model A is *nested* in model B if the explanatory variables of A are a subset of those of B. Let $\ell(\boldsymbol{\mu}; \boldsymbol{y})$ be the log-likelihood in terms of the mean vector $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_n)^\top$. The perfect fit is $\tilde{\boldsymbol{\mu}} = \boldsymbol{y}$, giving the maximum likelihood $\ell(\boldsymbol{y}; \boldsymbol{y})$: this is the *saturated model*, with a separate parameter for each observation. It does not smooth the data and is useless for predicting new data (overfitting); it is a *baseline* for comparison. Notation: $\tilde{\boldsymbol{\theta}}$ corresponds to $\tilde{\boldsymbol{\mu}} = \boldsymbol{y}$ (saturated); $\hat{\boldsymbol{\theta}}$ to the MLE $\hat{\boldsymbol{\mu}} = g^{-1}(\boldsymbol{X}\hat{\boldsymbol{\beta}})$ under the model. Always $\ell(\hat{\boldsymbol{\mu}}; \boldsymbol{y}) \le \ell(\boldsymbol{y}; \boldsymbol{y})$.

19.2 Scaled deviance and deviance

Definition 19.1. The *scaled deviance* of a GLM is twice the difference in maximum log-likelihood compared with the full (saturated) model.

$$D^*(\boldsymbol{y}; \hat{\boldsymbol{\mu}}) = -2[\ell(\hat{\boldsymbol{\mu}}; \boldsymbol{y}) - \ell(\boldsymbol{y}; \boldsymbol{y})] = \frac{2}{a(\phi)} \sum_{i=1}^{n} \Big\{ [y_i \tilde{\theta}_i - b(\tilde{\theta}_i)] - [y_i \hat{\theta}_i - b(\hat{\theta}_i)] \Big\}$$
$$= \frac{2}{a(\phi)} \sum_{i=1}^{n} \Big\{ y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i) \Big\} = \frac{D(\boldsymbol{y}; \hat{\boldsymbol{\mu}})}{a(\phi)}.$$

$$D(\boldsymbol{y}; \hat{\boldsymbol{\mu}}) = -2a(\phi)\,[\ell(\hat{\boldsymbol{\mu}}; \boldsymbol{y}) - \ell(\boldsymbol{y}; \boldsymbol{y})] \quad \text{(deviance)}, \qquad D^* = D/a(\phi) \quad \text{(scaled deviance)}$$

The deviance D does not contain $a(\phi)$ and is always computable; the scaled deviance D^* is computable only when ϕ is known. (The $c(y_i; \phi)$ terms cancel in the difference.)

Example 19.2 (The deviance generalizes RSS)**.** For the linear regression model $\theta = \mu = \eta$, $\phi = \sigma^2$, $a(\phi) = \phi$, $b(\theta) = \theta^2/2$, $c(y, \phi) = -\frac{1}{2}\big(\frac{y^2}{\phi} + \ln(2\pi\phi)\big)$. Saturated: $\tilde{\theta}_i = y_i$, so $y_i\tilde{\theta}_i - b(\tilde{\theta}_i) = y_i^2 - y_i^2/2 = y_i^2/2$. Model: $\hat{\theta}_i = \hat{\mu}_i$, so $y_i\hat{\theta}_i - b(\hat{\theta}_i) = y_i\hat{\mu}_i - \hat{\mu}_i^2/2$. Therefore

$$D^* = \frac{2}{\sigma^2} \sum_i \Big(\frac{y_i^2}{2} - y_i\hat{\mu}_i + \frac{\hat{\mu}_i^2}{2} \Big) = \frac{1}{\sigma^2} \sum_i (y_i - \hat{\mu}_i)^2 = \frac{\mathrm{RSS}}{\sigma^2}, \qquad D = \mathrm{RSS}\,.$$

Key insight: the deviance generalizes the residual sum of squares, the generalization being driven by the likelihood and its equivalence with RSS in the linear model.

Bernoulli deviance and the boundary limit. For $y_i \in \{0, 1\}$ with fitted $\hat{\pi}_i$,

$$D = 2 \sum_i \left[y_i \log \frac{y_i}{\hat{\pi}_i} + (1 - y_i) \log \frac{1 - y_i}{1 - \hat{\pi}_i} \right], \qquad 0 \log 0 = 0.$$

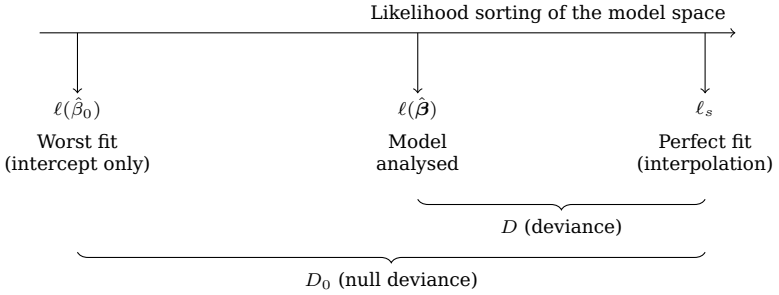
One cannot substitute the saturated natural parameter directly because $\mathrm{logit}(0) = -\infty$ and $\mathrm{logit}(1) = +\infty$. Take the combined limit in $y\theta - \log(1 + e^\theta)$: for $y = 1$ it tends to 0 as $\theta \to +\infty$; for $y = 0$ it tends to 0 as $\theta \to -\infty$. Hence the saturated Bernoulli log-likelihood is zero and $D = -2\ell(\hat{\pi})$ for ungrouped Bernoulli responses; check grouping and binomial constants before using the shortcut in general.

Poisson. $D = 2\sum_i \big[y_i \log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i) \big]$ with the limiting convention at $y_i = 0$.

19.3 Null deviance and the deviance R^2

Another benchmark is the *null deviance* D_0, the deviance of the model with only an intercept relative to the perfect model. It generalizes the total sum of squares: in the linear model

$$D_0 = \sum_{i=1}^{n} (y_i - \hat{\eta}_i)^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0)^2 = SST, \qquad \text{since } \hat{\beta}_0 = \bar{y} \text{ with no predictors.}$$



Using D and D_0 one compares how much the model improves by adding $x_1, \ldots, x_p$, the *percentage of deviance explained*, through the generalization of the coefficient of determination

$$R^2 \quad := \quad 1 \; - \; \frac{D}{D_0}$$

Larger deviance indicates poorer goodness of fit. Deviance-based hypothesis tests (differences of deviances of nested models as likelihood ratio statistics) belong to the inference chapter.

Six-point logistic example. For $x = (0, 1, 2, 3, 4, 5)$ and $y = (0, 0, 1, 1, 0, 1)$, the logistic fit has coefficients approximately $\hat{\beta}_0 = -1.6841$, $\hat{\beta}_1 = 0.67365$ and Bernoulli deviance about 6.77284. The identities $\sum \hat{\pi}_i = \sum y_i$ and $\sum x_i \hat{\pi}_i = \sum x_i y_i$ follow from the score equations for an intercept and one slope under the canonical logit link; they are practical checks that do not hold without qualification for every link. R: `glm(..., family=binomial())`, `logLik()`, `deviance()`, `predict(type="response")`.

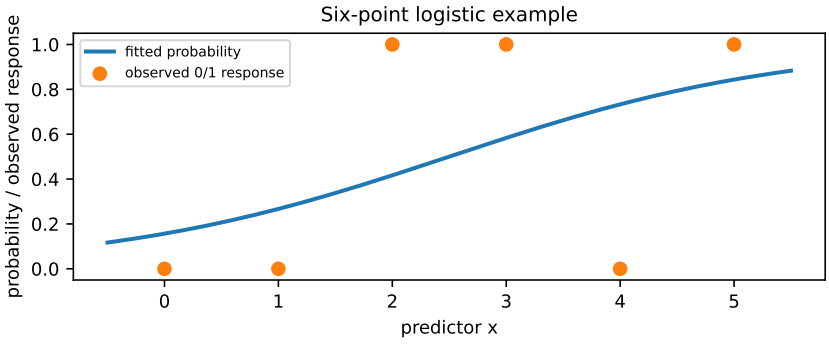


Figure 4: Six-point Bernoulli-logit curve and observed outcomes, drawn with the reported coefficients.

20 Logistic regression in practice

Instead of linear OLS regression we use logistic regression, which (with its variants) is probably more widely used than OLS. **Why not SLR?** (1)

← **Proof idea**
Distinguish a model for the conditional mean from the noise model.

SLR may predict values below 0 or above 1. (2) SLR assumes a normal distribution around the mean for each value of X, which cannot hold for a binary response.

Testing the model. The likelihood ratio test compares the maximum of the likelihood function for each (nested) model; the deviance difference is the LRT statistic.

Goodness of fit. The Hosmer–Lemeshow test is the most widely used goodness-of-fit test for logistic regression. Variable and model selection can be combined with cross-validation (Section 22).

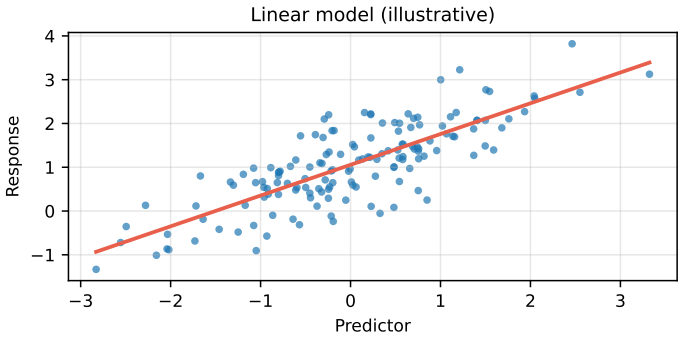


Figure 5: Output from code example 1 (illustrative reconstruction). R packages: ISLR2, ISLR; functions: `plot`, `glm`, `predict`.

21 Family-by-family reference

Model families
I check $V(\mu)$ first, then link, scale, domain and weights.

Normal / identity. $\theta = \mu$, $V(\mu) = 1$, $g(\mu) = \mu$. The IRLS weight is constant and the solution reduces to OLS; the variance scale is still needed for exact normal-model uncertainty statements.

Bernoulli / logit and binomial / logit. $\mu = \pi$, $\theta = \log[\pi/(1-\pi)]$, $V(\mu) = \mu(1-\mu)$, canonical link logit. Binomial data can be successes out of m trials or proportions with case weights; track which version the software uses. If $Y \sim \mathrm{Bin}(m, \pi)$ is a count, $EY = m\pi$ and $\mathrm{Var}\, Y = m\pi(1-\pi)$; state the link in terms of π or μ to avoid mixing them.

Poisson / log. $\theta = \log \mu$, $V(\mu) = \mu$, $g(\mu) = \log \mu$, $W_{ii} = \mu_i$. Log-link Poisson regression keeps all means positive. Exposure enters as an offset such as $\log t_i$; coefficients are then multiplicative rate ratios.

Exponential and Gamma / inverse or log. For an exponential observation with mean μ the natural parameter is $-1/\mu$ and $V(\mu) = \mu^2$. A Gamma model with fixed shape has the same $V(\mu) = \mu^2$ up to dispersion. The canonical link is inverse (with a sign convention); the log link is a common alternative, and with a log link $g'(\mu) = 1/\mu$ so the IRLS weight is constant when $V(\mu) = \mu^2$.

Inverse Gaussian / inverse-square or log. $V(\mu) = \mu^3$, canonical link proportional to $-1/\mu^2$. With a log link $g'(\mu) = 1/\mu$ and $W_{ii} = 1/\mu_i$ (up to dispersion). The mean domain restriction matters: a numerical update must not generate an invalid prediction.

Family	Domain	$V(\mu)$	Canonical link	Alternative
Gaussian	real	1	identity	log (positive means)
Bernoulli	$(0,1)$	$\mu(1-\mu)$	logit	probit, cloglog
Poisson	positive	μ	log	identity with care
Gamma	positive	μ^2	inverse	log
Inverse Gaussian	positive	μ^3	inverse-square	log

The family controls the variance function; the link controls the path from

μ to η. Choose them separately, then inspect deviance residuals and the plausibility of the assumptions. R: `gaussian()`, `binomial()`, `poisson()`, `Gamma()`, `inverse.gaussian()`, `family()`, `glm.fit()`, `residuals(type="deviance")`.

22 Cross-validation: predicting on observations not used to fit

Cross-validation

Validation requires held-out predictions and refitting preprocessing within each training fold.

Three errors not to confuse. *Training error* is the average loss on the observations used to fit. *Test-set error* uses separate held-out observations. *Expected test error* is the expected loss over future independent data and ordinarily cannot be observed directly. Cross-validation estimates predictive performance by repeatedly holding observations out.

The K-fold estimator. Partition the data into folds $I_1, \dots, I_K$ and let $\hat{f}^{(-k)}$ be fitted after excluding fold I_k. For a loss L,

$$\widehat{\mathrm{Err}}_{\mathrm{CV}} = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in I_k} L\{y_i, \hat{f}^{(-k)}(x_i)\}$$

Variable selection, imputation, standardization or tuning done on the whole dataset before making folds leaks validation data into training; those operations belong inside each training fold.

LOOCV, K-fold, repeated and nested CV. Leave-one-out uses $K = n$; it can be expensive, though linear models have a shortcut. Five- or ten-fold CV balances training fraction and cost. Repeated K-fold reveals instability across partitions. Nested CV is needed when tuning and also assessing the whole tuning procedure: inner folds choose settings, outer folds estimate held-out performance.

The LOOCV shortcut for linear regression. For OLS with a fixed full-rank design,

$$e_i^{(-i)} = \frac{e_i}{1 - h_{ii}}, \qquad \mathrm{LOOCV} = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{e_i}{1 - h_{ii}} \right)^2,$$

the delete-one identity of Proposition 8.5 reappearing in out-of-sample prediction.

Choosing the loss. Squared or absolute error for continuous responses; log loss for probabilistic predictions; misclassification error depends on a threshold and ignores calibration; Poisson deviance can be more natural for counts than raw squared error. Write down the loss before fitting.

When random folds are inappropriate. Observations from the same patient, household, hospital or time series may not be independent; random folds then leak information and are over-optimistic. Grouped CV, blocked CV or rolling-origin validation may match the real prediction setting. R: `sample()`, `split()`, `predict()`, `glm()`, `lm()`, `hatvalues()`, `boot::cv.glm()`, `rsample::vfold_cv()`.

23 Proof-checking list and summary questions

Last connections

Geometry, inference and numerical algorithms are describing the same fitting process.

1. **Before invoking a distribution:** which assumptions give the result? Projection geometry needs fewer assumptions than exact t/F sampling distributions.
2. **Before differentiating a GLM likelihood:** is the parameter θ, μ, η or β? Write every derivative in the chain.

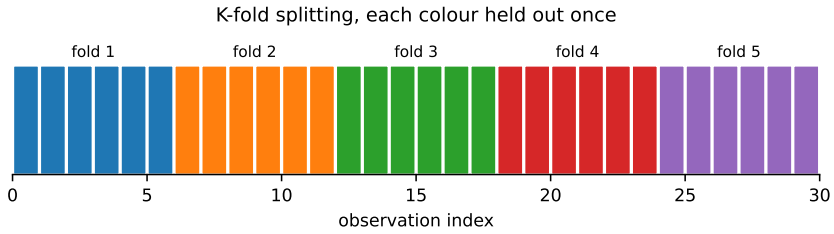


Figure 6: Illustrative fold assignments and the held-out predictions idea. R: `sample()`, `split()`, `rsample::vfold_cv()`, `ggplot2::geom_tile()`.

3. **Before calling a link canonical:** does it identify η with the natural parameter under the actual exponential-family parameterization (h is the identity)?
4. **Before doing IRLS:** how are $\boldsymbol{W}$ and the working response $\boldsymbol{\xi}$ obtained? Do they agree with the score and Fisher information?
5. **Before using a deviance:** which saturated model, constants, dispersion conventions and zero-count limits are involved? Is it D or D^*?
6. **Before comparing AIC or BIC:** are the likelihoods comparable and the fitted datasets identical?
7. **Before interpreting CV:** were preprocessing, feature selection and tuning performed inside the training folds?
8. **Before interpreting numerical output:** do rank, separation, extreme weights, influence and the response domain undermine the computation?

Summary questions (Chapter 5). How to determine an MLE for a GLM? What are the three algorithms? How are the three linked (Newton–Raphson uses $\mathcal{J}$; Fisher scoring replaces it by $\mathcal{I}$; IRLS is Fisher scoring written as a weighted least-squares update)? If the canonical link is applied, what result follows (h is the identity, the score is $\boldsymbol{X}^\top(\boldsymbol{y}-\boldsymbol{\mu})/a(\phi)$, and Fisher scoring equals Newton–Raphson)? What is the mathematical definition of the (scaled) deviance? How is the GLM deviance a generalization of the linear-model RSS, and the null deviance of the SST?

The broader idea to keep: **geometry, probability and computation are three views of one modeling process**. The projection picture comes from linear algebra, likelihood from mathematical statistics, and numerical analysis matters the moment the model is actually fitted.